

# Robot Learning

General course information  
Basics of robotics  
Fundamentals of machine learning

# Team



Prof. Erdem Bıyık  
Instructor  
[biyik@usc.edu](mailto:biyik@usc.edu)

# Office hours

- Erdem:
  - Friday, 9:00am – 10:00am, GCS SB3 (Floor: LL2)

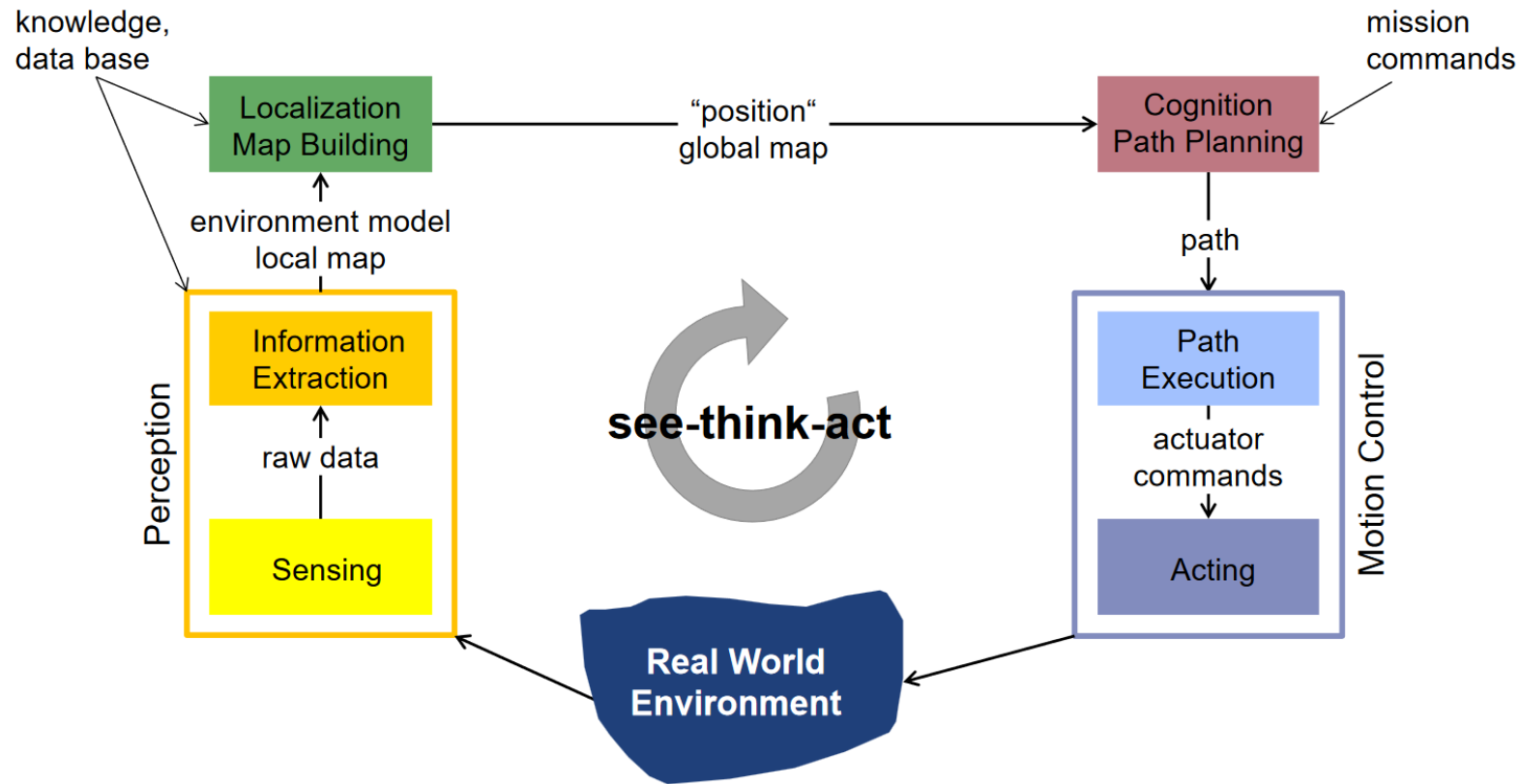
# Online resources

- Course website: <https://liralab.usc.edu/csci699/>
- Piazza: <https://piazza.com/usc/fall2025/csci699specialtopics>
- Gradescope: <https://www.gradescope.com/courses/1073531>

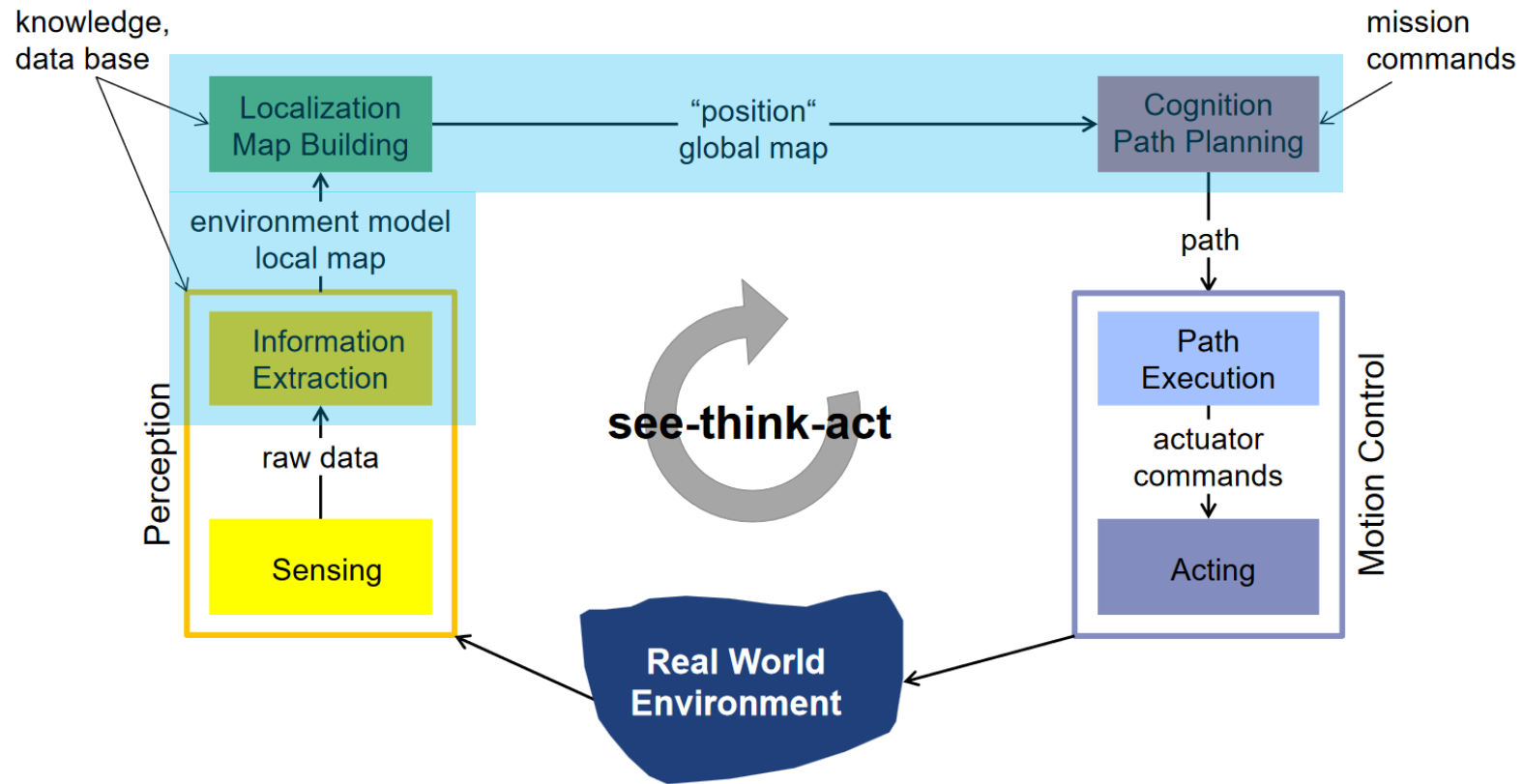
# What is a robot?

- An embodied artificial intelligence
- A machine that can autonomously carry out useful work
- An artificial device that can sense its environment and purposefully act on or in that environment

# See-think-act cycle



# Robot learning



# Why robot learning?

## Designing controllers is hard

- Requires good understanding of the system
- Doesn't scale well to high-dimensional systems
- *“Manipulation breaks all the rigorous/reliable approaches I know for control.”*
  - Russ Tedrake (MIT / TRI)

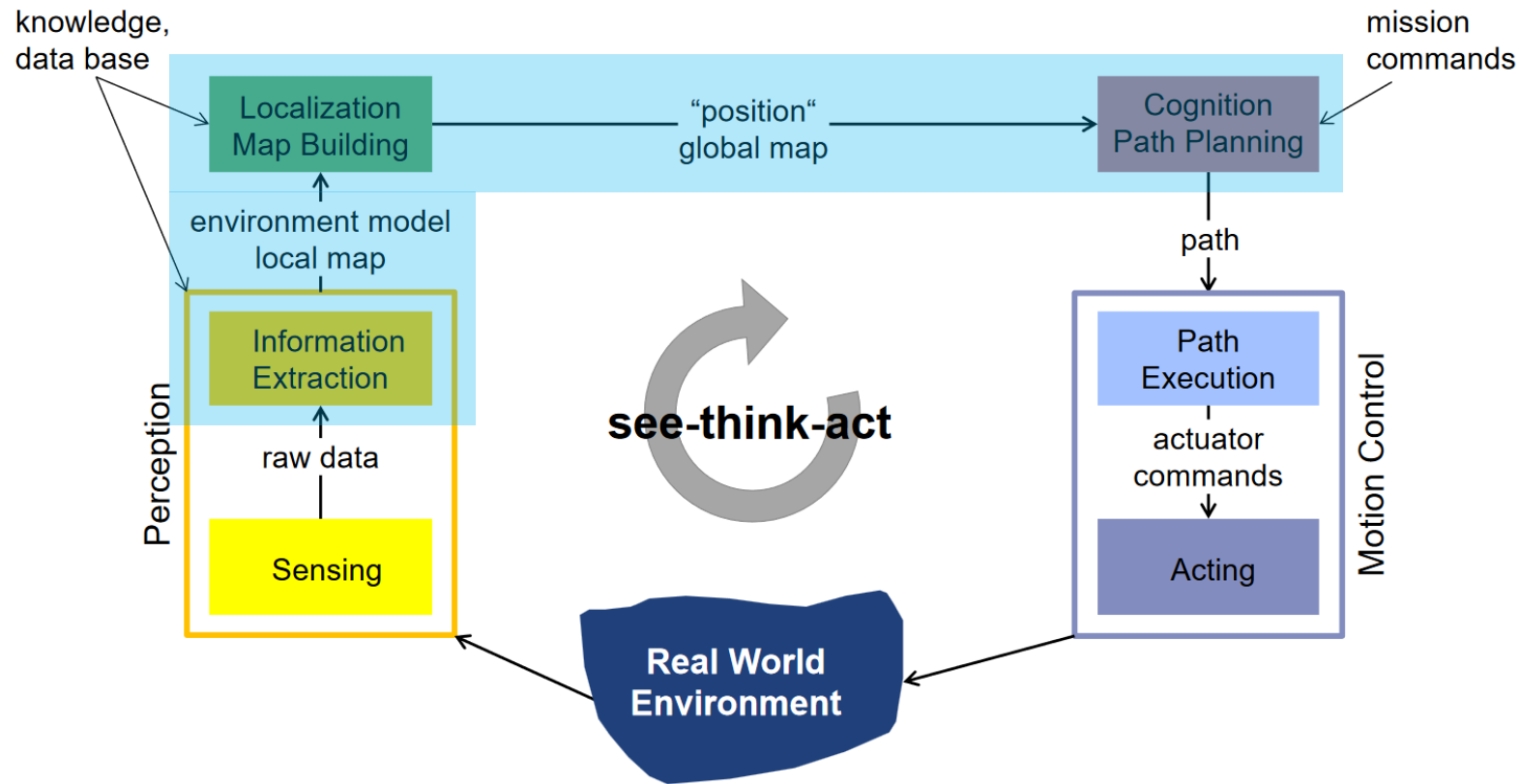
# Prerequisites

- Probability theory
- Calculus
- Linear algebra
- At least one programming language (preferably, Python)
  - Programming assignments will be in Python.
- Recommended:
  - Familiarity with basic concepts in machine learning

# What's covered?

- Basics of...
  - Robotics
  - Machine learning
  - Computer vision
- Representation learning
- Reinforcement learning
- Imitation learning / IRL
- Learning from human feedback
- Sim-to-real transfer
- Meta-learning
- Safe and robust learning
- Multi-agent learning
- Robot learning using natural language

# What's NOT covered?



# What's NOT covered?

- Robot operating system (ROS)
  - CSCI 545: Robotics
- Simultaneous localization and mapping (SLAM)
  - CSCI 545: Robotics
- Grasping and manipulation
  - CSCI 699: Deep Learning for Robotic Manipulation

# Textbook & Readings

- No textbook is required.
- All readings will be available on course website.
- If I were to recommend textbooks for this class...
  - Reinforcement Learning: An Introduction by Sutton and Barto
  - Modern Adaptive Control and Reinforcement Learning (MACRL) by Bagnell, Boots, and Choudhury
  - Principles of Robot Autonomy by Lorenzetti and Pavone

# Assignments

- One class project (40%)
- Two homework assignments (2 x 15%)
- Three paper presentations (3 x 10%)

# Homework assignments

- Three homework assignments in total – you choose which two you would like to do!
  - Basics of robotics & machine learning & computer vision
  - Reinforcement learning
  - Imitation learning & intent inference & shared autonomy
- Both theoretical and programming components
- Programming parts will be in Python
- No ROS knowledge required
- The submissions will be online, due at 12 midnight
- Each student has 3 free late days.

# Homework assignments

- The class does not have a TA or a grader.
- They told me to be creative about how to handle grading.
- After every deadline, I will share the solution key with you.
- Everyone grades themselves in two weeks.
- I randomly pick  $n$  submissions and grade them myself.
  - If I realize the student cheated by increasing their score by  $x$ , I give a penalty of  $kx/n$  points where  $k$  is the number of students who submitted the homework.
- I repeat until I find  $n$  fairly graded assignments.

# Class presentation

- 15-25 minutes presentation, depending on the week/paper
- Should include an extensive discussion of the paper
  - Motivation
  - Prior work
  - Methods
  - Results
  - Discussion
    - Both the positive and the negative aspects of the paper!
- 5 minutes Q&A

# Course project

- The project will be done in groups of 2 or 3.
- Feel free to reach out to me if you have a good reason to do it individually or as a group of more than 3 students.

# Course project

The project must have both robotics and machine learning components.

Examples:

- Application-dependent improvements over an existing robot learning method
- A new application of an existing robot learning technique
- A novel method that **may** have potential benefits

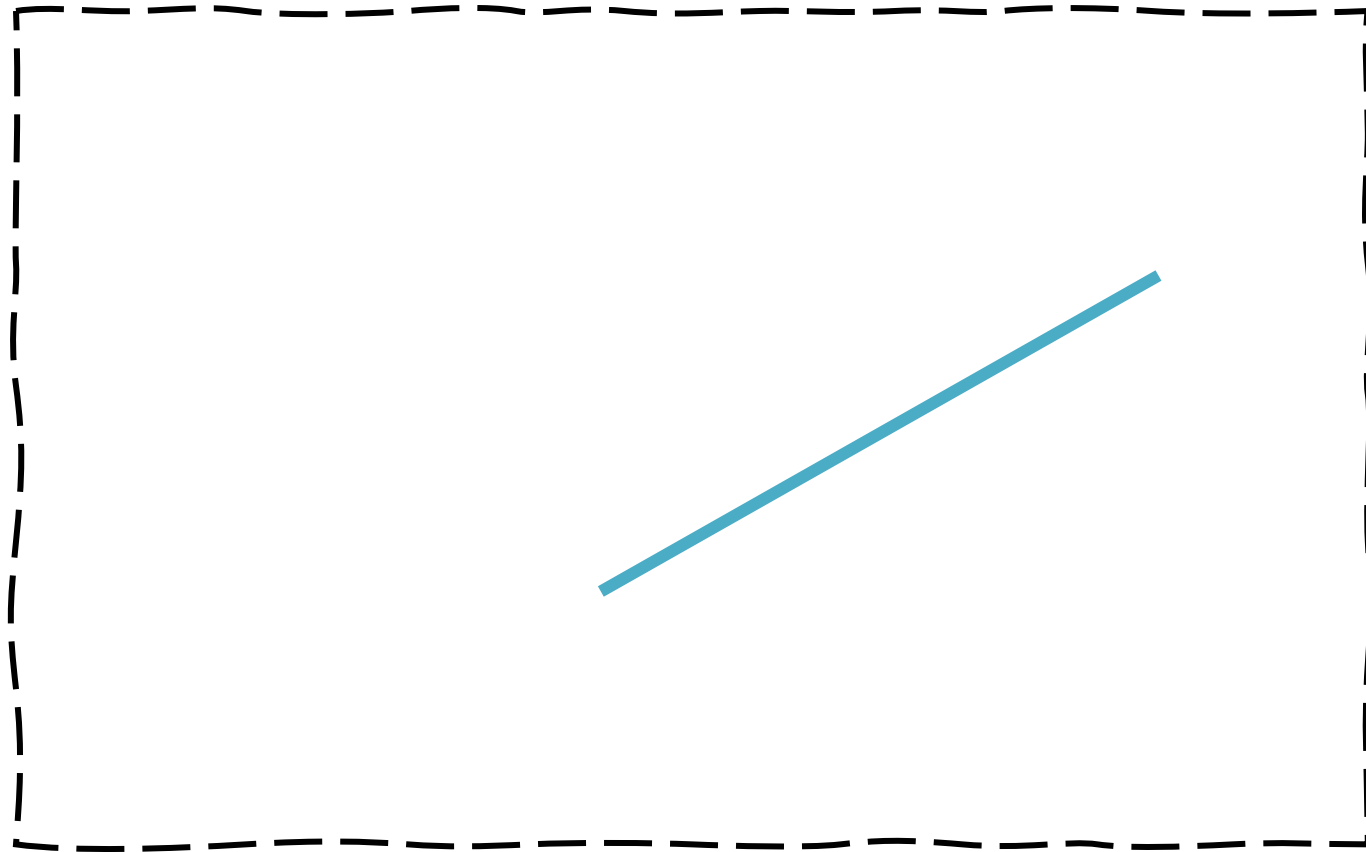
# Course project

| Component                                 | Contribution to Grade |                           |
|-------------------------------------------|-----------------------|---------------------------|
| Project Proposal Report                   | 5%                    | October 19 <sup>th</sup>  |
| Project Milestone Report                  | 5%                    | November 16 <sup>th</sup> |
| Project Presentation (Possibly with Demo) | 10%                   | December 5 <sup>th</sup>  |
| Final Project Report                      | 15%                   | December 7 <sup>th</sup>  |
| Peer Review                               | 5%                    | December 14 <sup>th</sup> |
| Total                                     | 40%                   |                           |

# Today...

- General course information
- Basics of robotics
- Fundamentals of machine learning

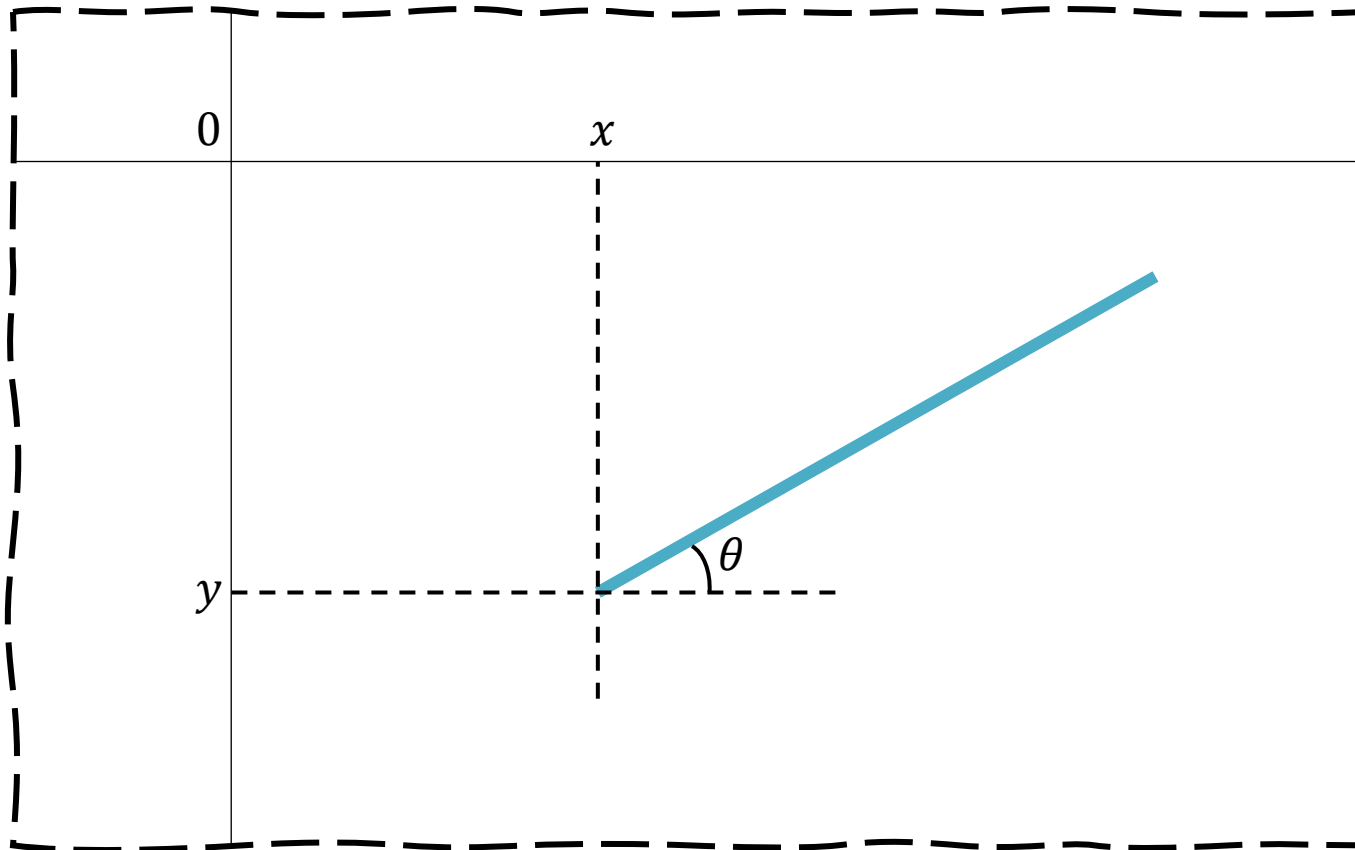
# A rigid body in 2D space



This rigid body is free to move and rotate in any direction.

How many variables do we need to fully describe the configuration of this rigid body?

# A rigid body in 2D space

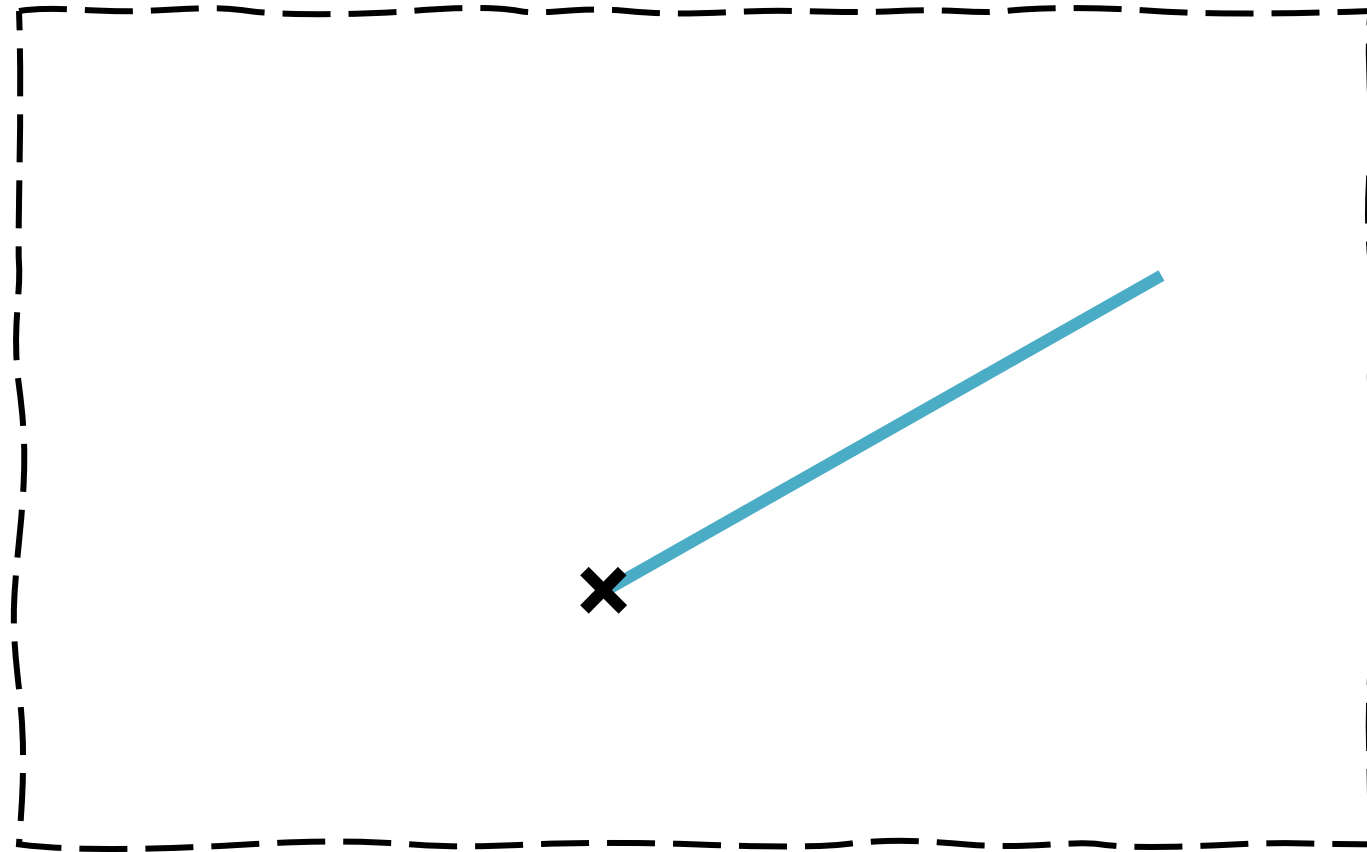


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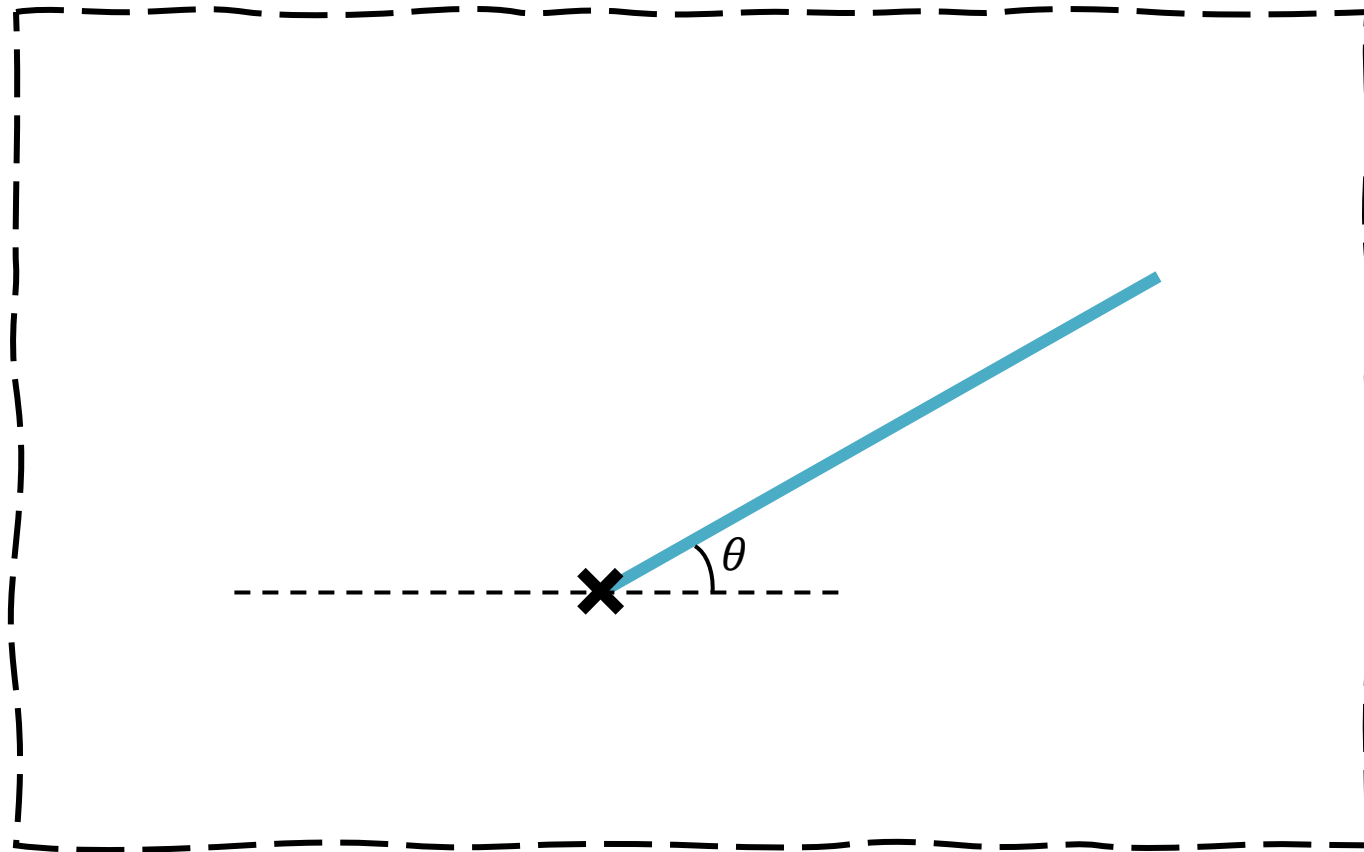
The answer is 3 variables:  $(x, y, \theta)$

# A rigid body in 2D space



What if one of the end points is fixed?

# A rigid body in 2D space



What if one of the end points is fixed?

Two of the variables are now fixed by two constraints:

$$\begin{aligned}x &= \bar{x} \\ y &= \bar{y}\end{aligned}$$

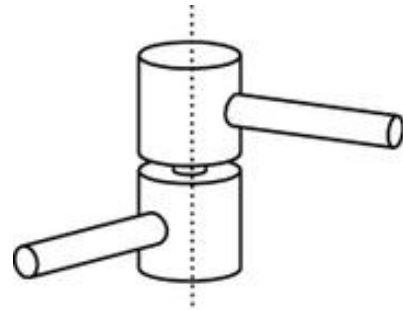
We only need one variable:  $\theta$

This is called the **degree-of-freedom (DoF)** of the body.

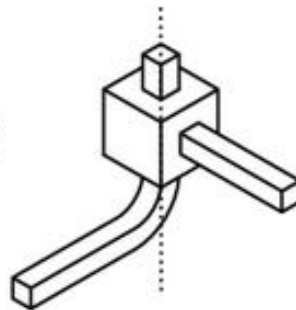
# A rigid body in 3D space

- Requires 6 degrees of freedom:
  - Three for position
  - Three for orientation

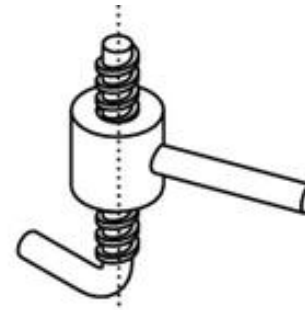
# Common joints



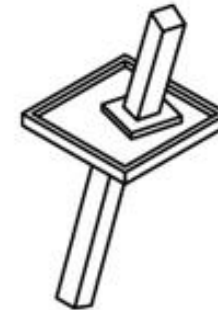
Revolute joint



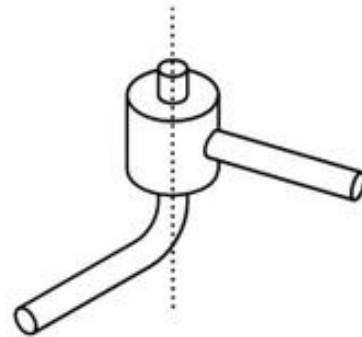
Prismatic joint



Helical joint



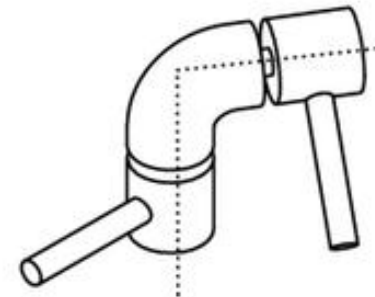
Planar joint



Cylindrical joint

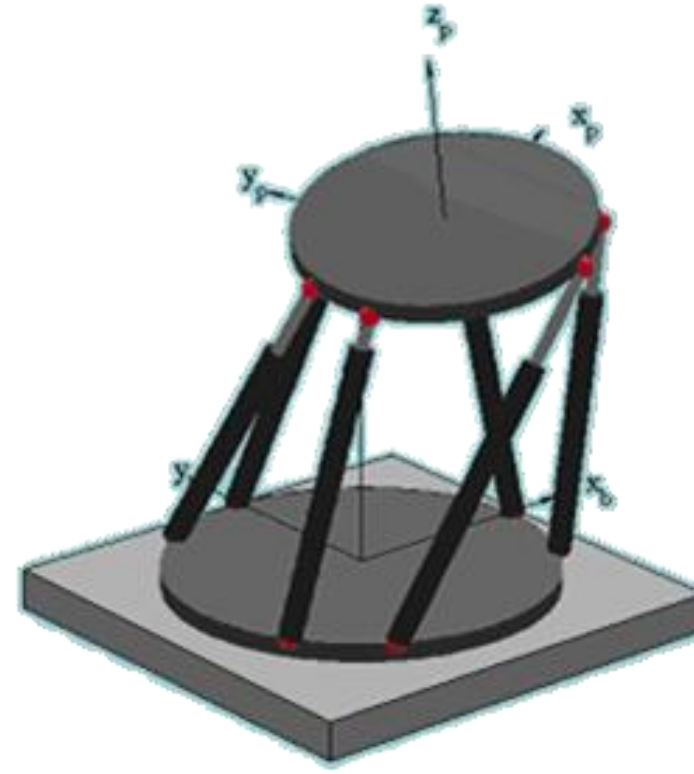
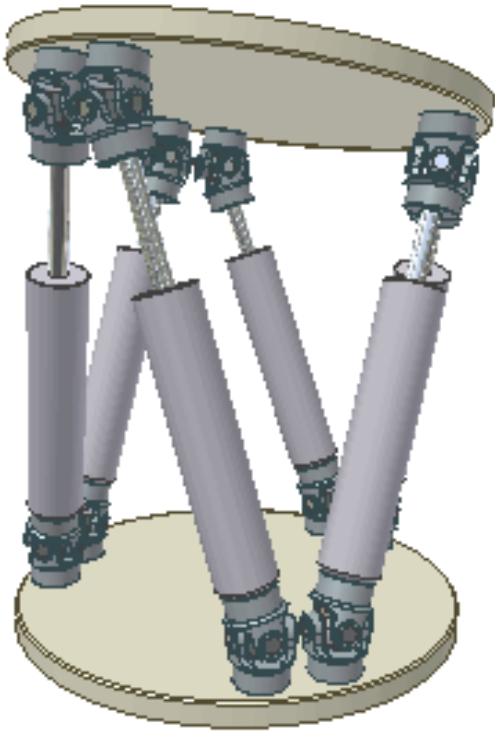


Spherical joint



Universal joint

# Degrees of freedom of a robot



# Grübler's formula

$N$  = # of bodies (including ground)

$J$  = # of joints

$$m = \begin{cases} 3, & \text{if planar} \\ 6, & \text{if spatial} \end{cases}$$

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$$\text{dof} = m(N - 1) - \sum_{i=1}^J c_i$$

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Number of independent joint constraints

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$$\begin{aligned} \text{dof} &= m(N - 1) - \sum_{i=1}^J c_i \\ &= m(N - 1) - \sum_{i=1}^J (m - f_i) \end{aligned}$$

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$$\begin{aligned} \text{dof} &= m(N - 1) - \sum_{i=1}^J c_i \\ &= m(N - 1) - \sum_{i=1}^J (m - f_i) = m(N - 1 - J) + \sum_{i=1}^J f_i \end{aligned}$$

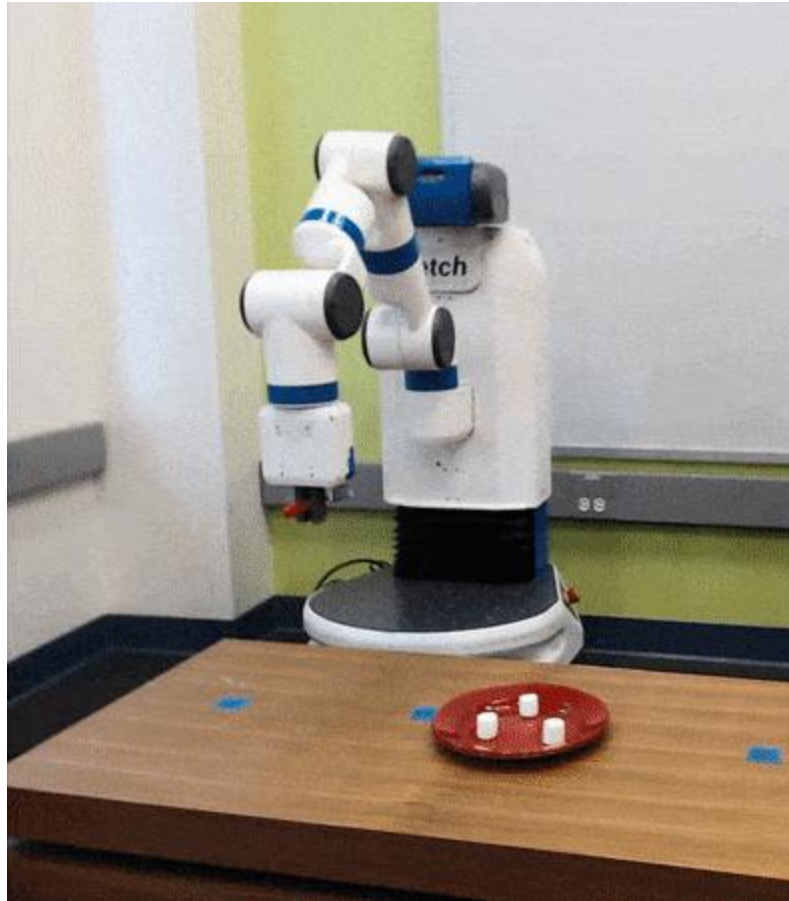
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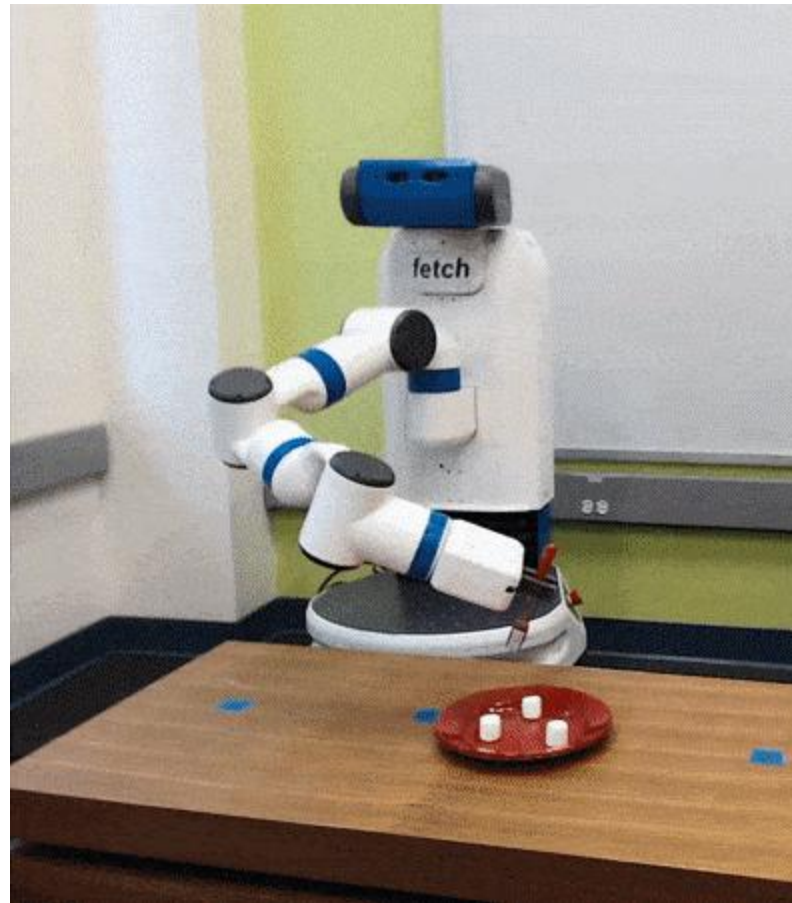
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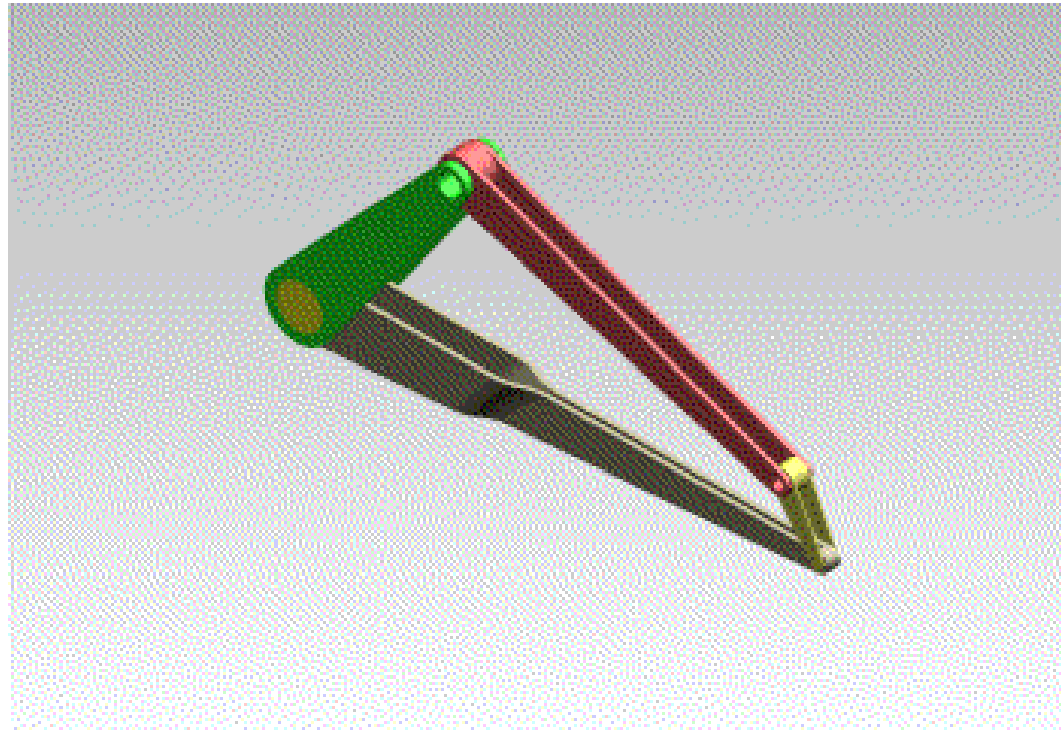
# An open-chain robot arm



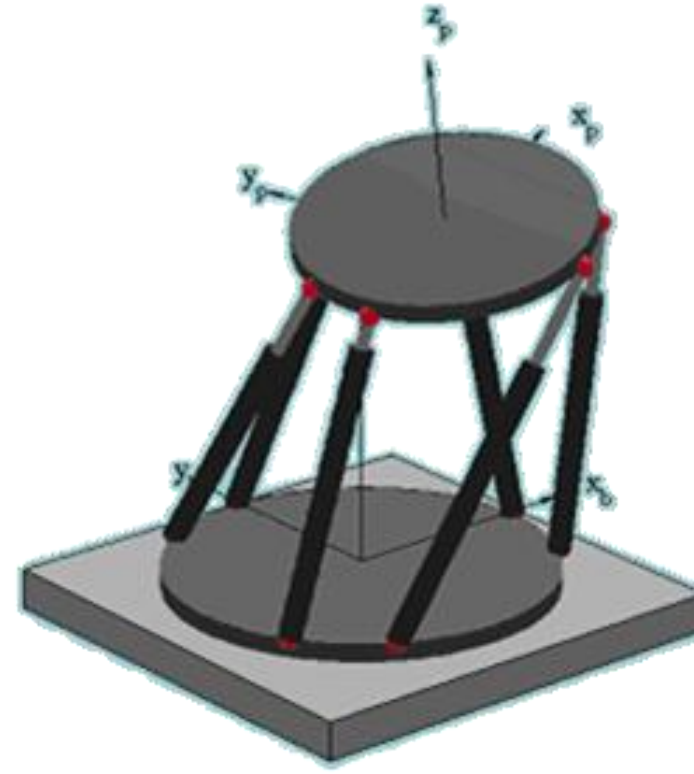
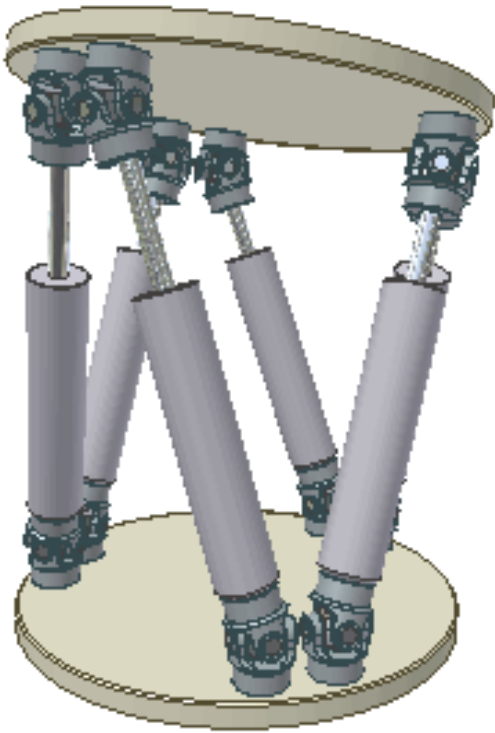
# An open-chain robot arm



# Four-bar closed-chain mechanism



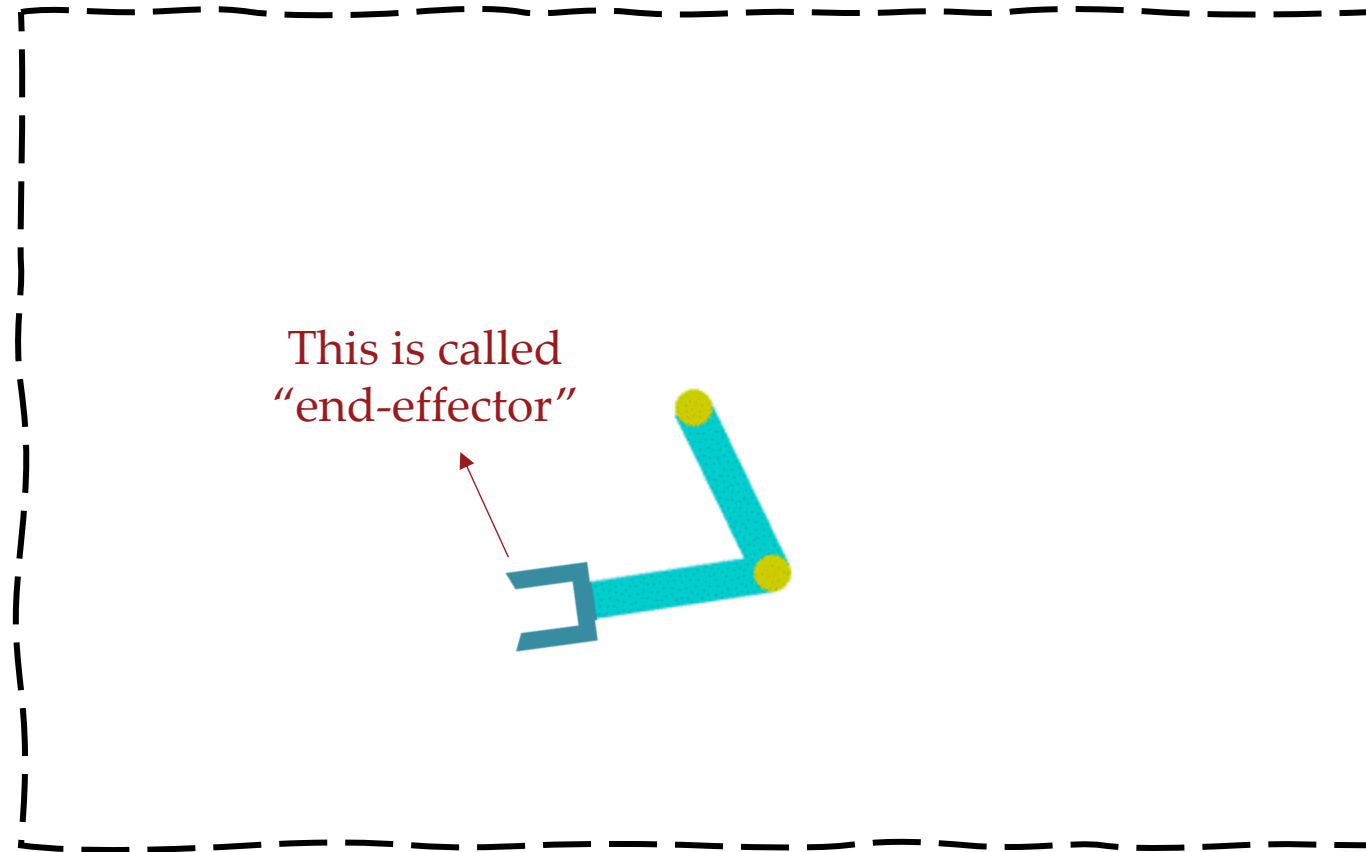
# Stewart platform



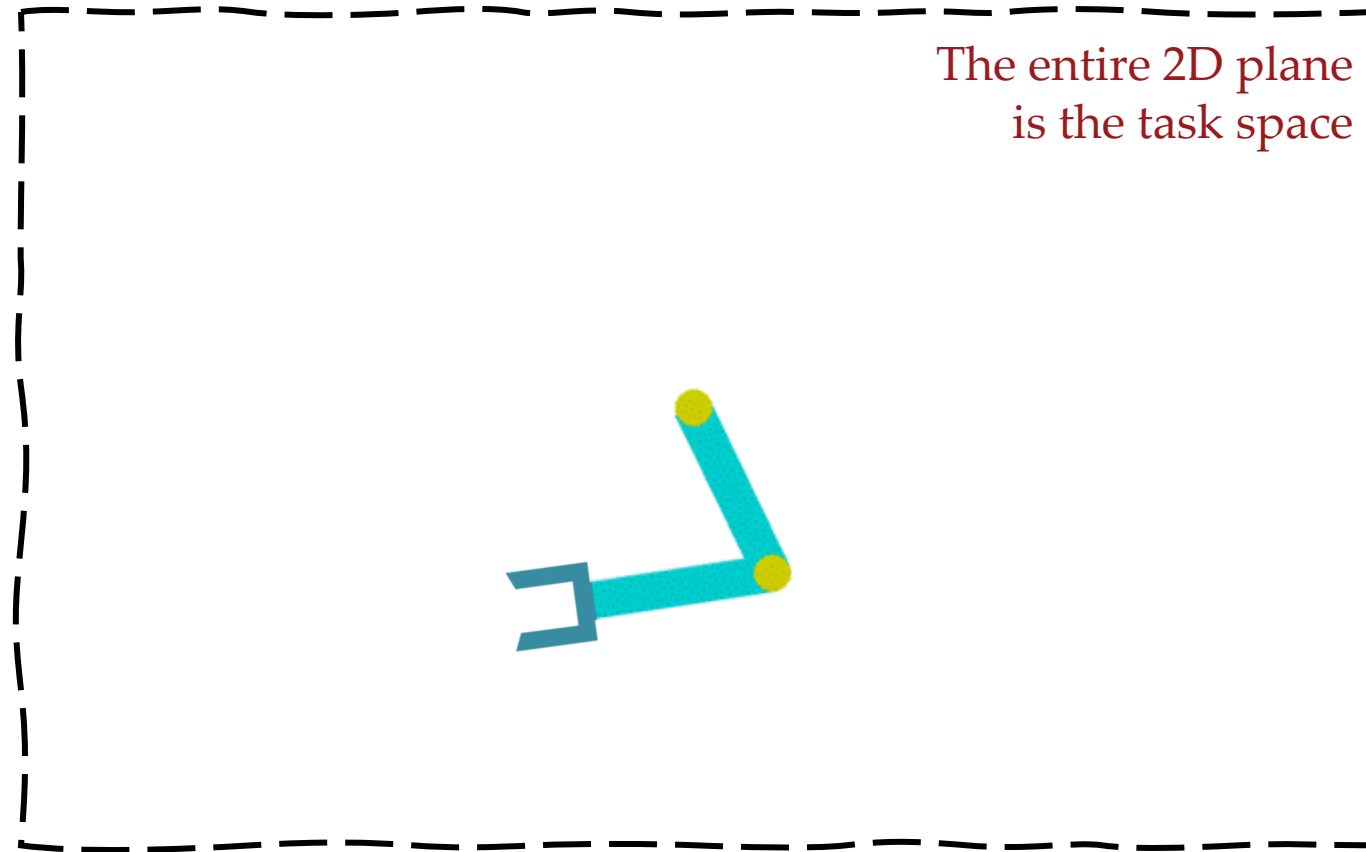
# Acrobot



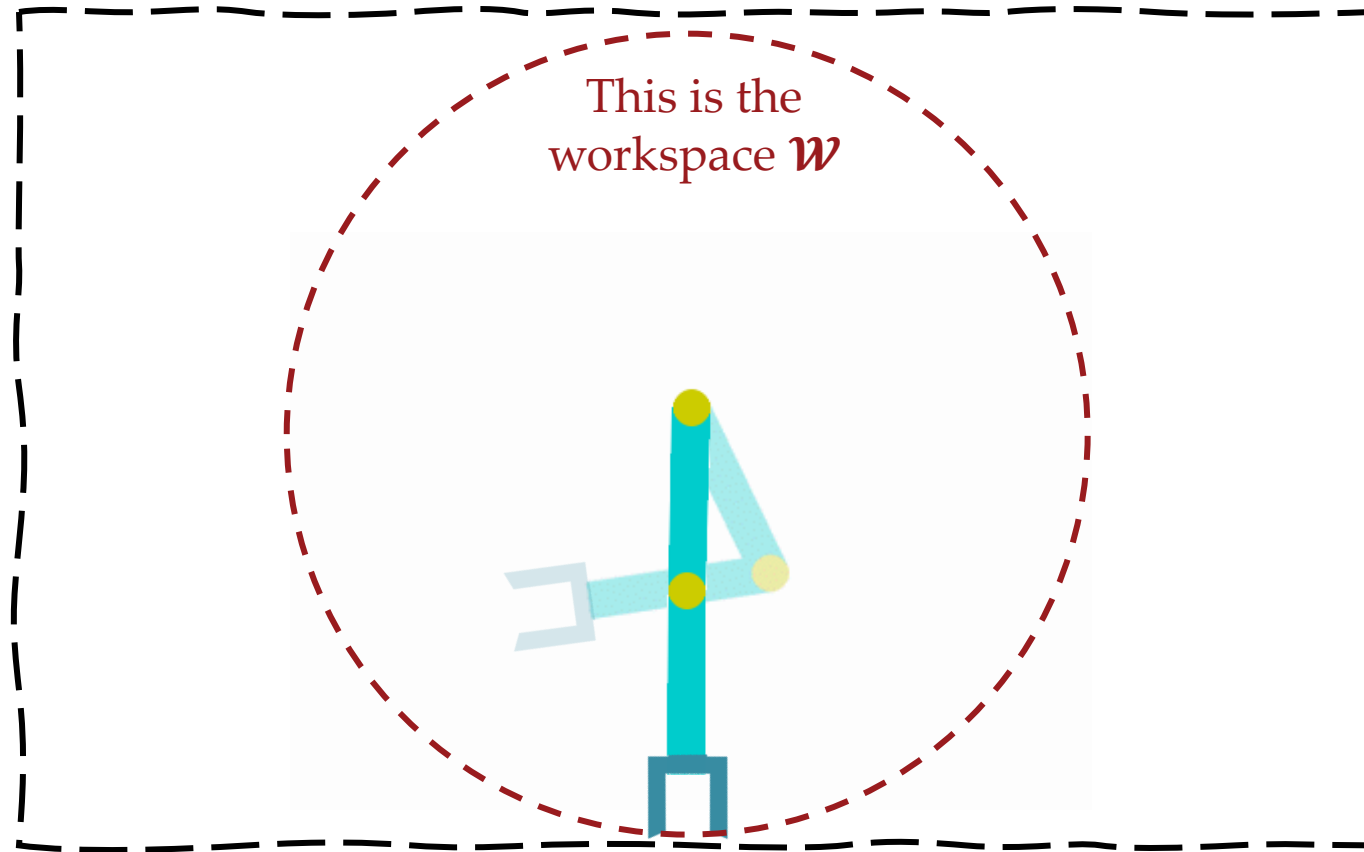
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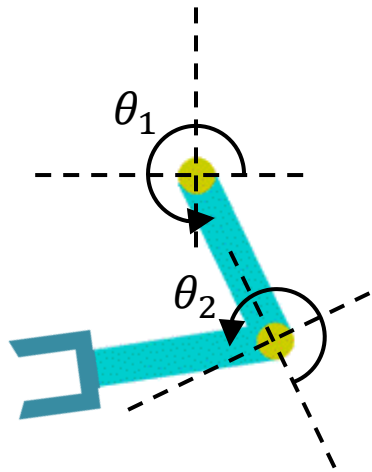
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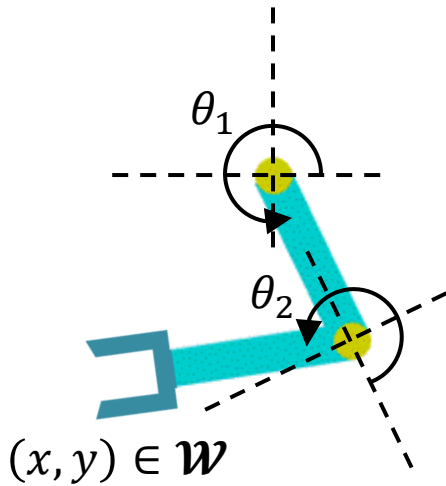


The robot's current configuration is:

$$(\theta_1, \theta_2) \in \mathcal{C}$$

This is called the  
configuration space

# Acrobot



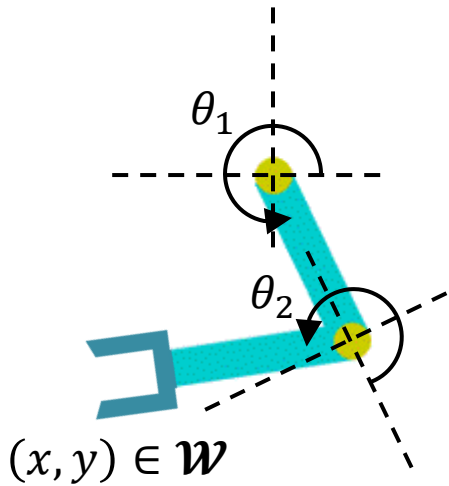
The robot's current configuration is:

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Forward kinematics:  $\text{FK}: \mathcal{C} \rightarrow \mathcal{W}$

$$\text{FK}((\theta_1, \theta_2)) = (x, y)$$

# Acrobot



The robot's current configuration is:

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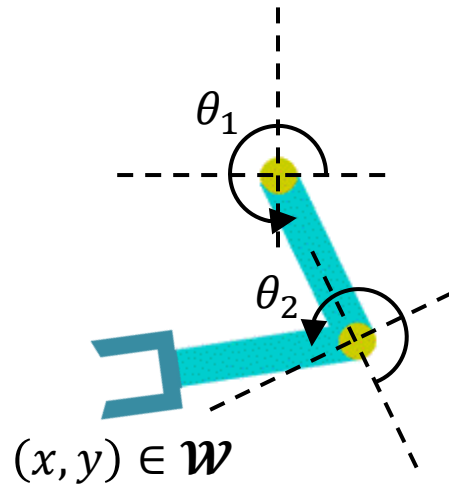
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Inverse kinematics: IK:  $\mathcal{W} \rightarrow \mathcal{C}$

$$\text{IK}((x, y)) = (\theta_1, \theta_2)$$

# Acrobot



The robot's current configuration is:

$$(\theta_1, \theta_2) \in \mathcal{C}$$

Forward kinematics:  $\text{FK}: \mathcal{C} \rightarrow \mathcal{W}$

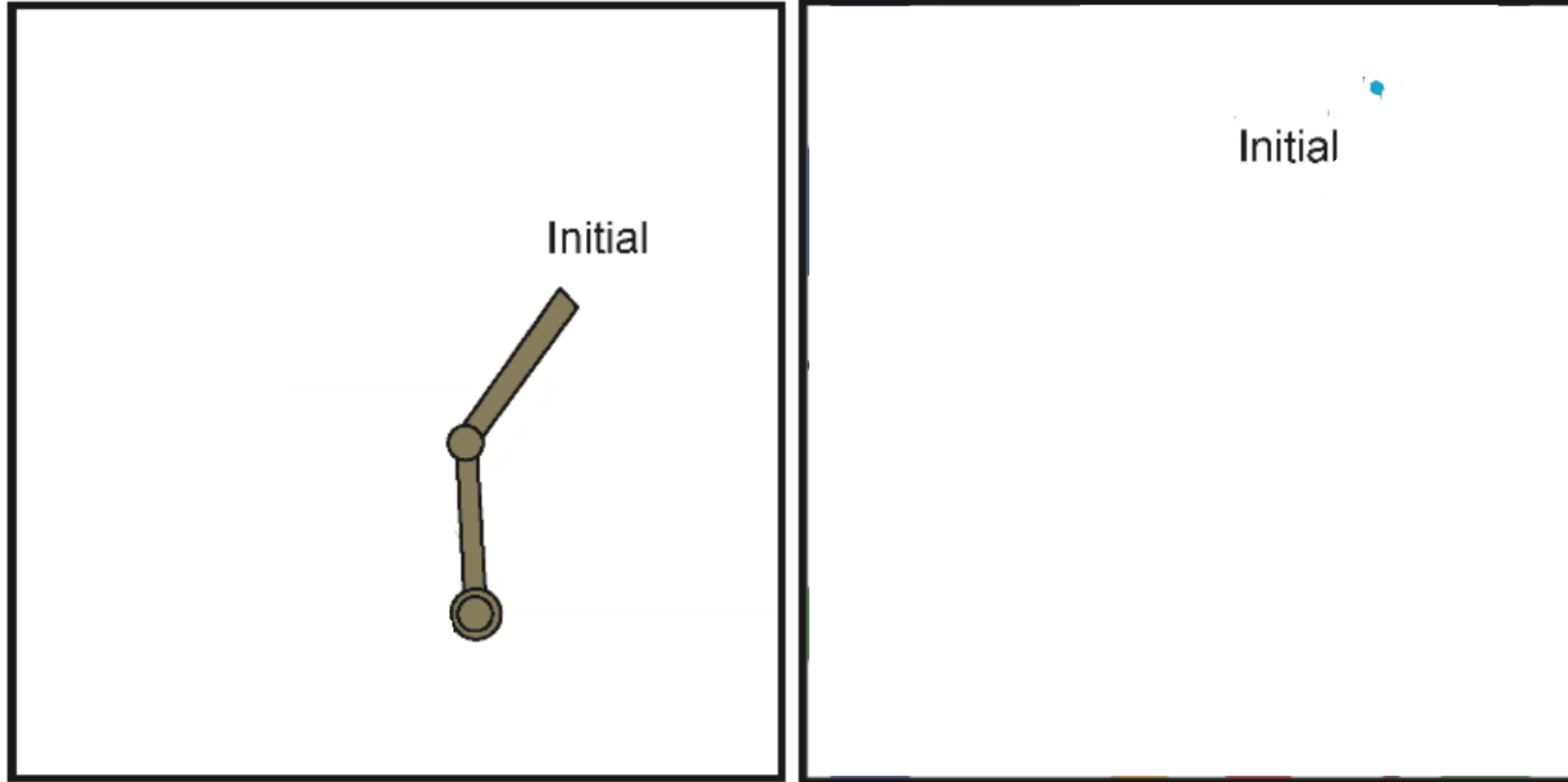
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Inverse kinematics:  $\text{IK}: \mathcal{W} \rightarrow \mathcal{C}$

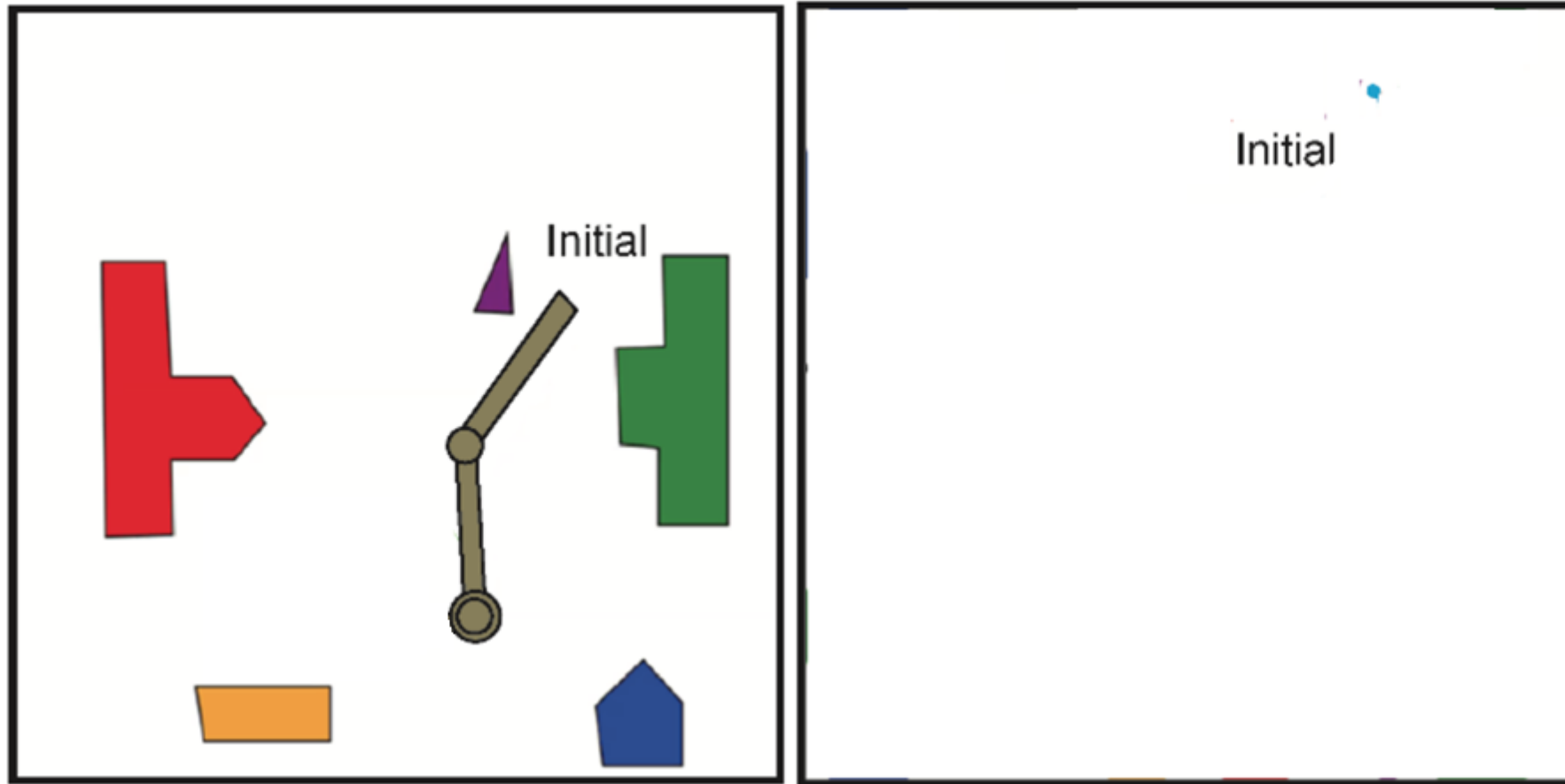
$$\text{IK}((x, y)) = (\theta_1, \theta_2)$$

This is often not a proper function.  
Because many configurations may  
lead to the same end-effector pose.

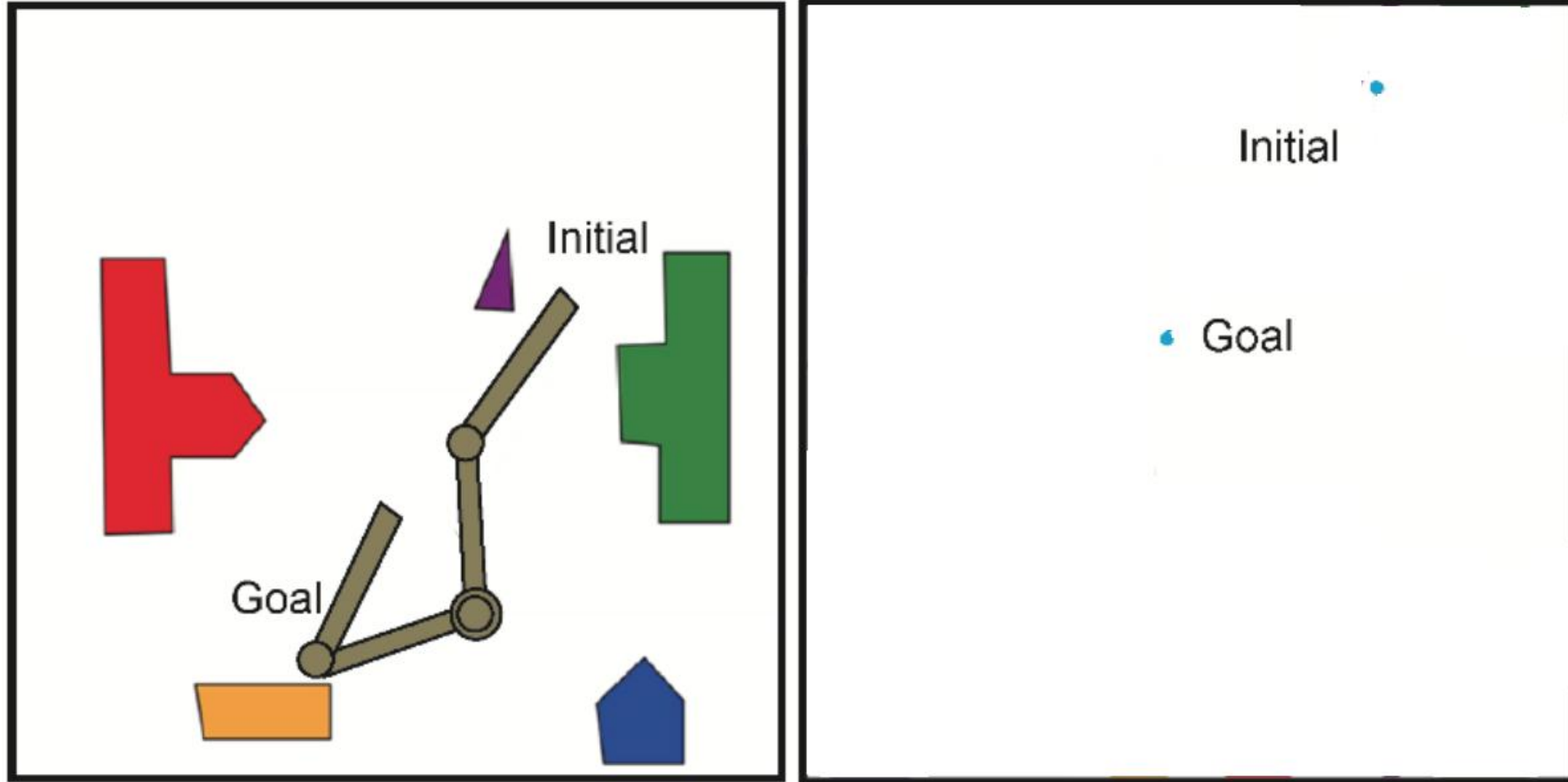
# Okay, but why?



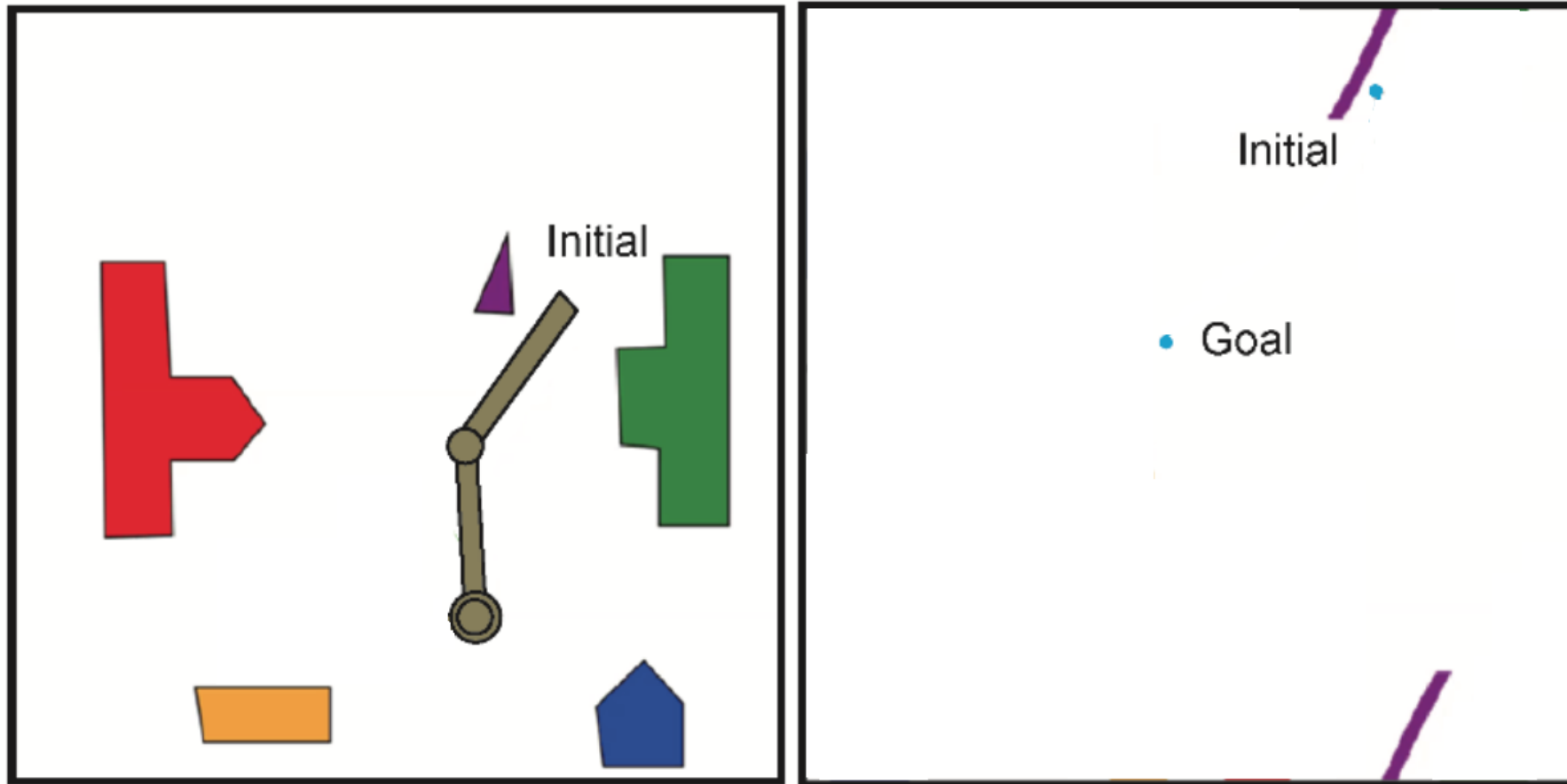
# Okay, but why?



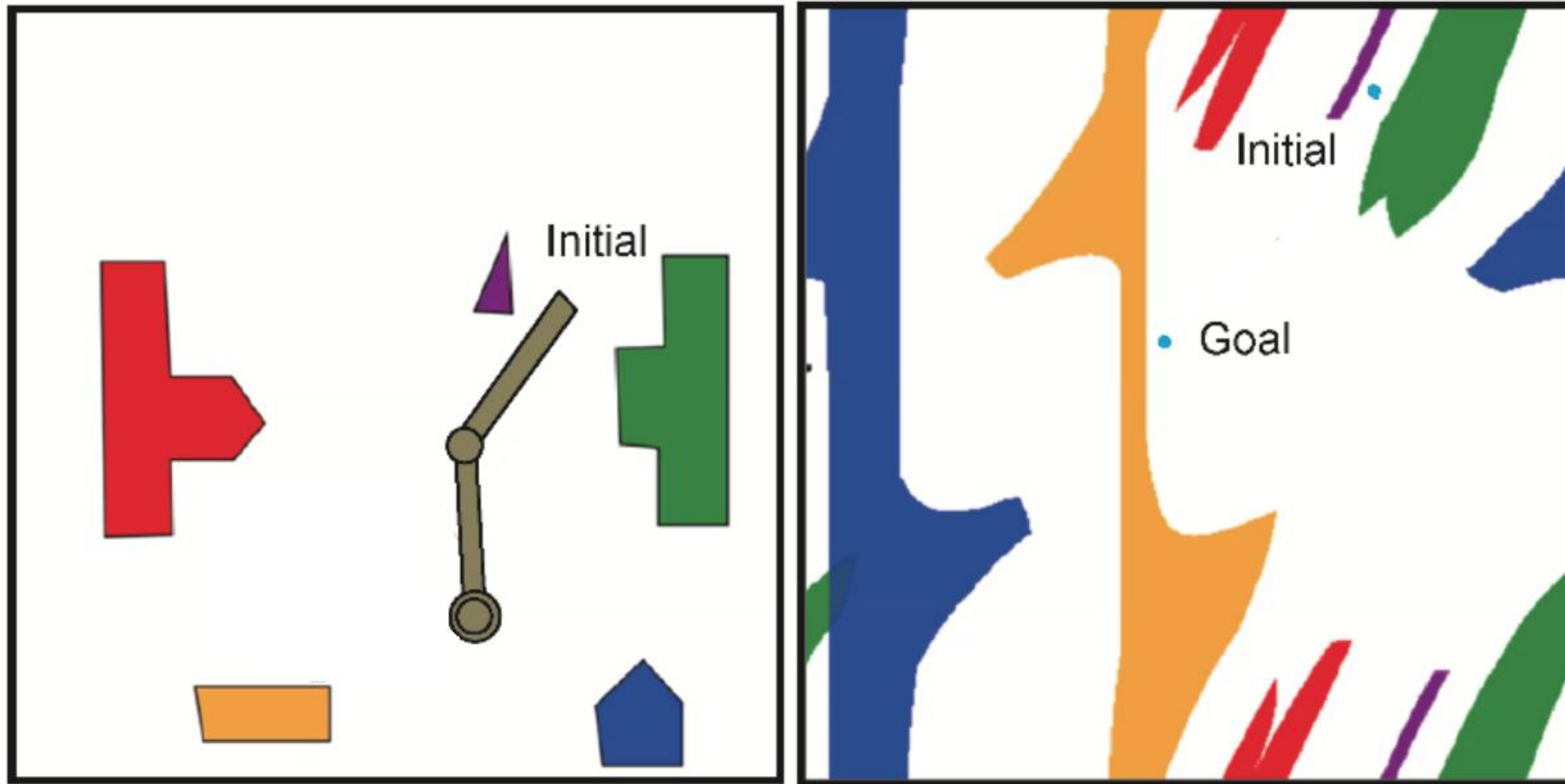
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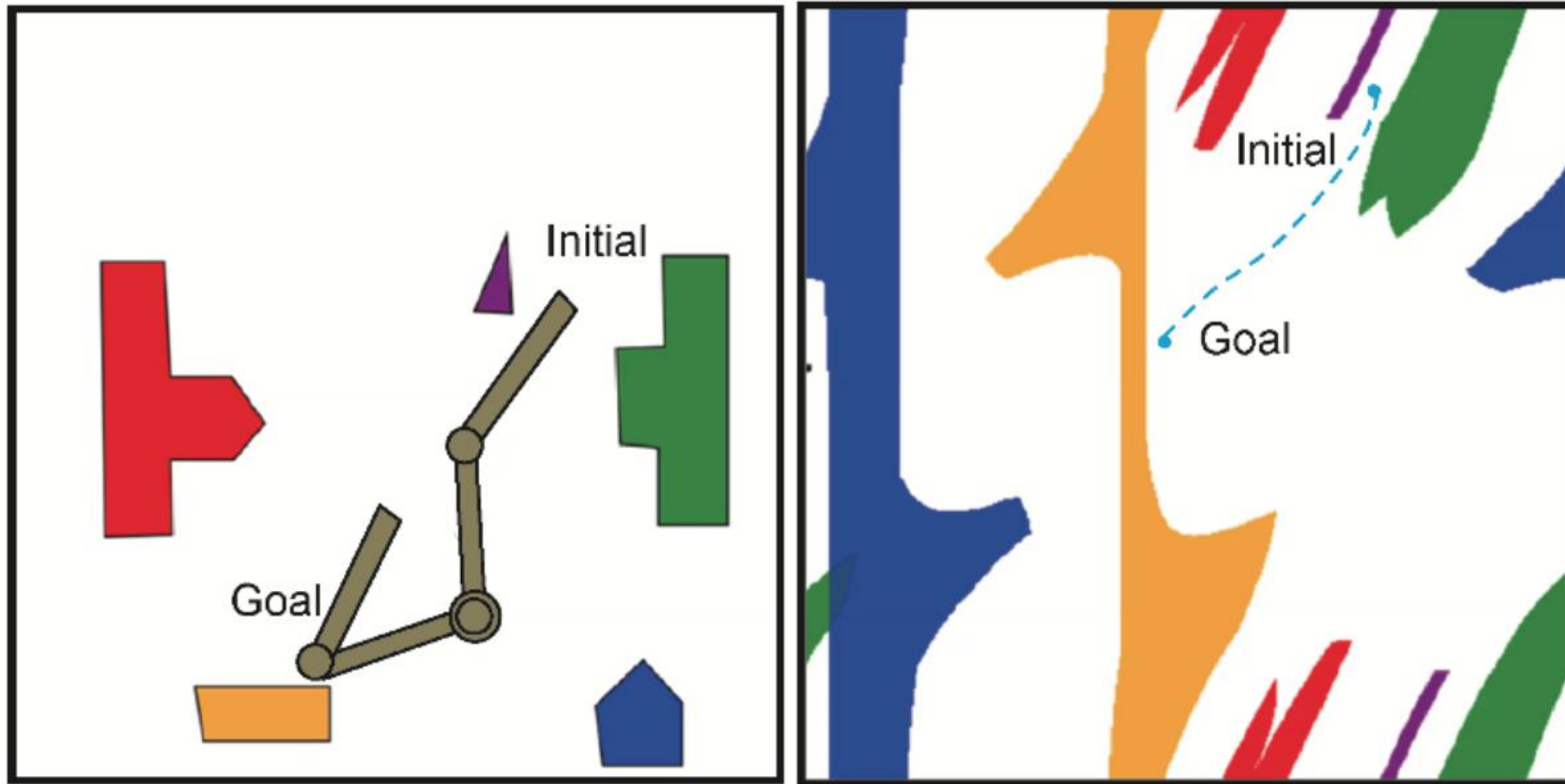
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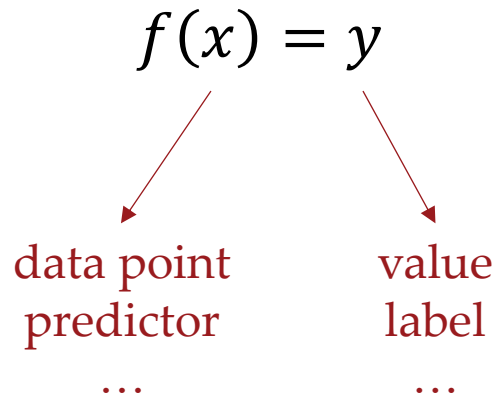
# Today...

- General course information
- Basics of robotics
- Fundamentals of machine learning

# Machine learning

## Supervised learning

Given  $\{(x^i, y^i)\}_{i=1}^n$ , find a function



(classification, regression)

## Unsupervised learning

# Machine learning

## Supervised learning

Given  $\{(x^i, y^i)\}_{i=1}^n$ , find a function

$$f(x) = y$$

data point  
predictor

...

value  
label

...

(classification, regression)

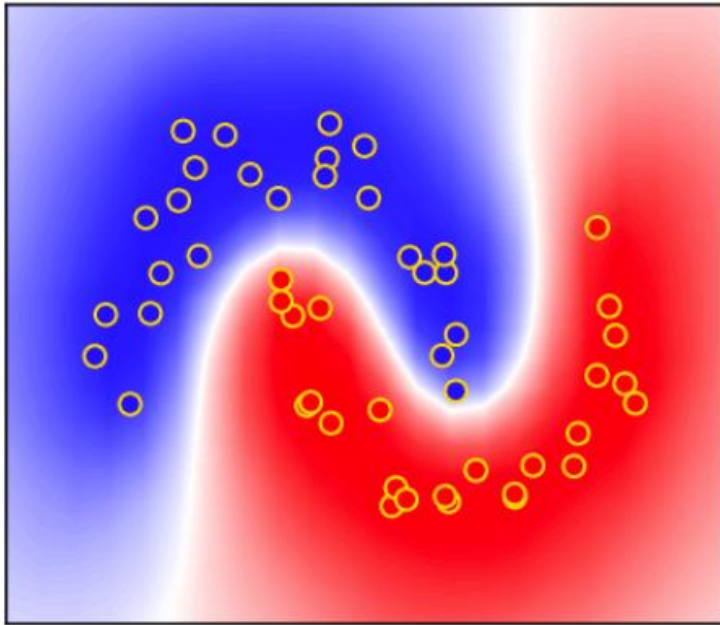
## Unsupervised learning

Given  $\{x^i\}_{i=1}^n$ , find patterns

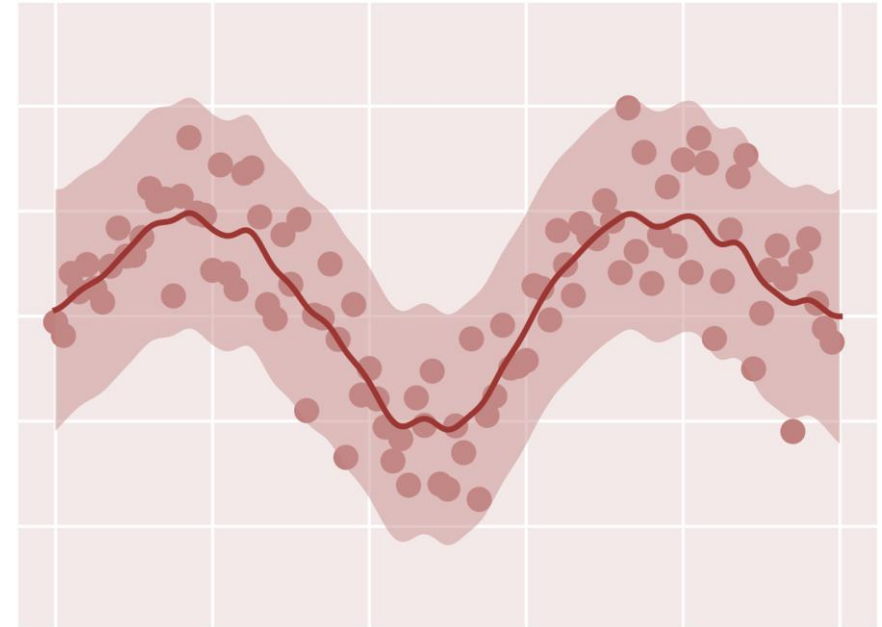
(clustering, compression,  
dimensionality reduction)

# Supervised learning

## Classification



## Unsupervised learning



# Learning models

- Parametric models:

$$y = f_{\theta}(x)$$

Examples: naïve Bayes, logistic regression, neural networks

# Learning models

- Parametric models:

$$y = f_{\theta}(x)$$

Examples: naïve Bayes, logistic regression, neural networks

- Non-parametric models:

$$y = f(x; D)$$

dataset



Examples: K-nearest neighbors, Gaussian process regression

# Loss functions

A loss function evaluates the quality of fit in  $f(x) \approx y$  or the quality of patterns in an unsupervised learning problem.

Examples:

$\ell^2$  loss: 
$$L(\theta) = \sum_{(x^i, y^i) \in D} (y^i - f_{\theta}(x^i))^2$$

Cross-entropy loss: 
$$L(\theta) = - \sum_{(x^i, y^i) \in D} (y^i)^{\top} \log f_{\theta}(x^i)$$

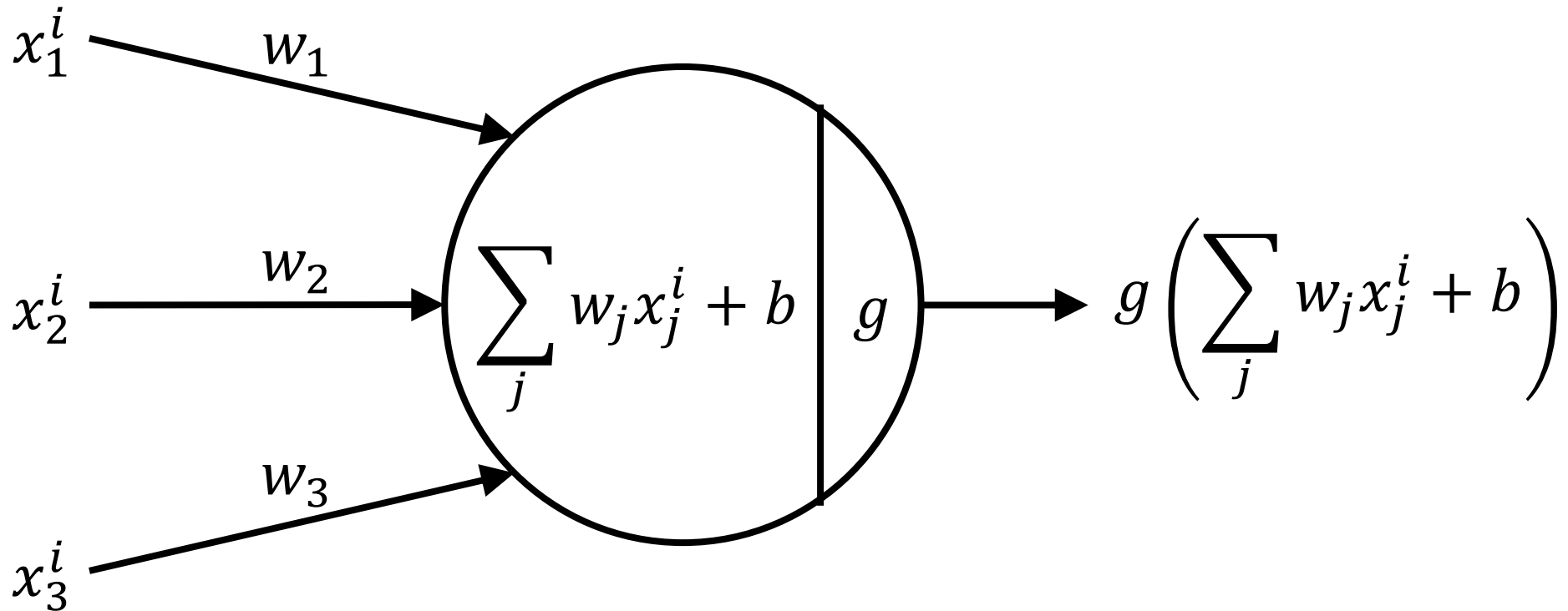
# Minimizing the loss

- Analytical solution
  - Use exact methods to find  $\theta^* = \arg \min_{\theta} L(\theta)$
  - Occasionally possible, e.g., linear regression
- Numerical optimization
  - Numerically minimize  $L(\theta)$ , e.g., gradient descent by computing  $\nabla L(\theta)$
  - Much more common in robot learning research
  - Stochastic optimization is often necessary for efficiency

# Neural networks

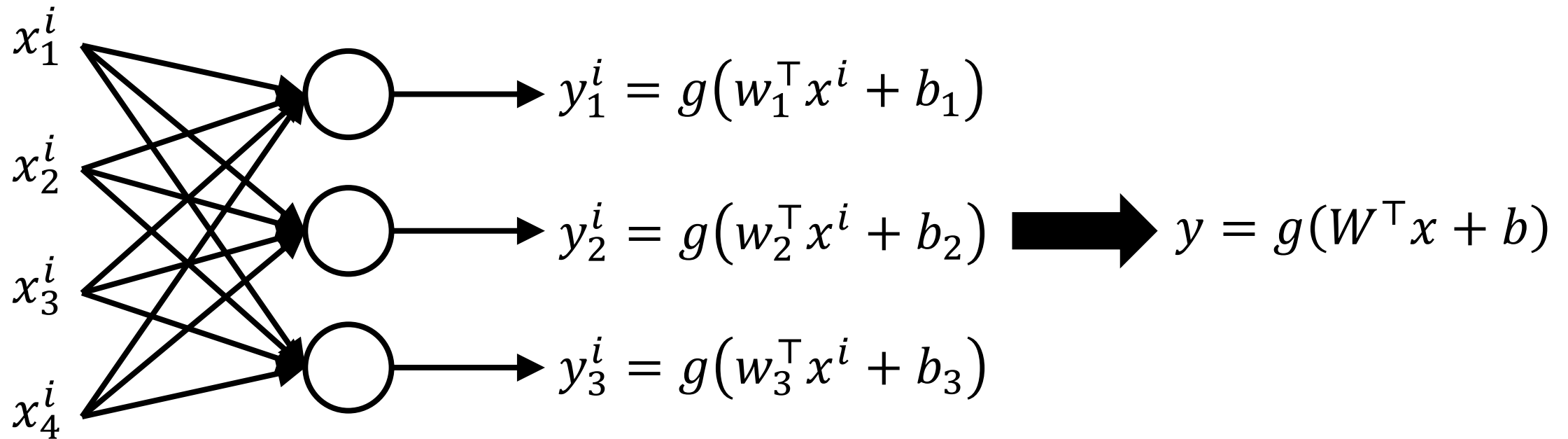
This is not the first model taught in a machine learning class. But we will almost never use other models.

## 1. A perceptron



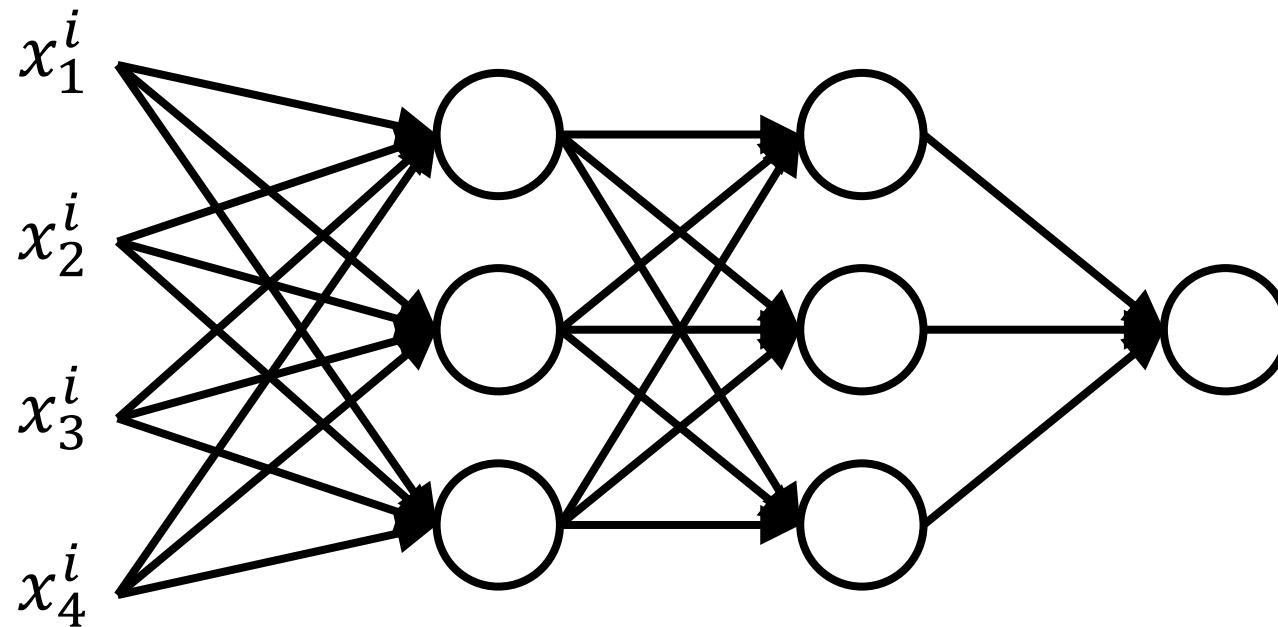
# Neural networks

## 2. A single layer neural network

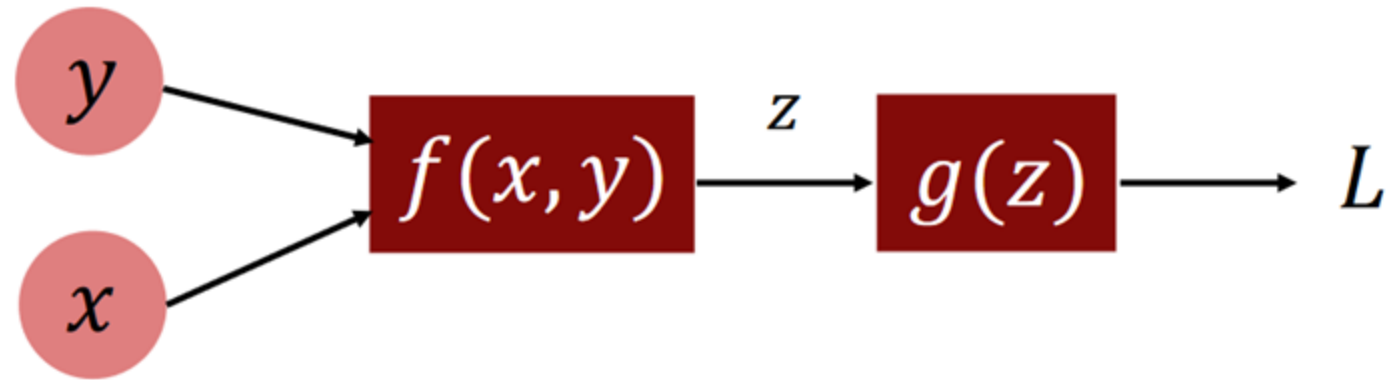


# Neural networks

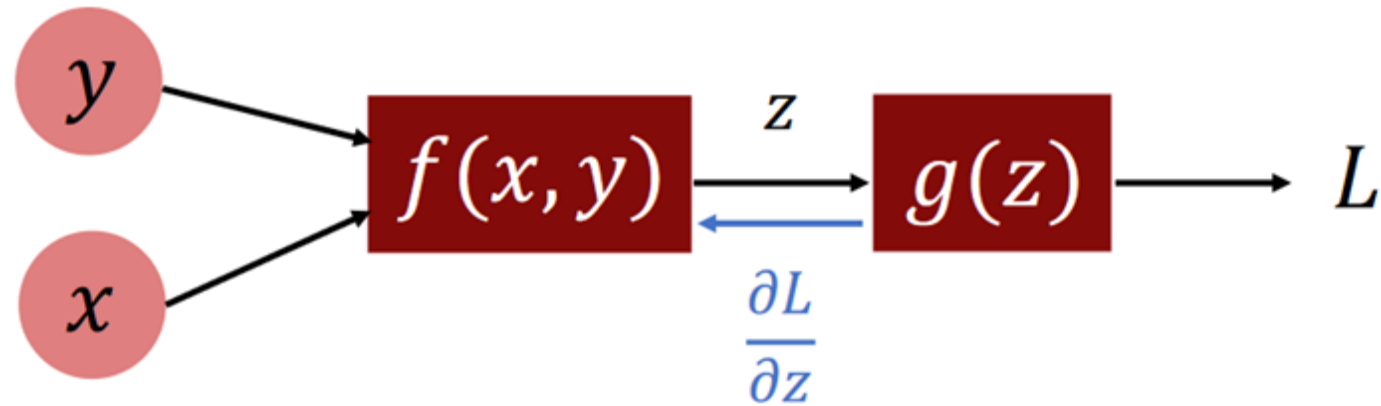
## 3. A deep neural network



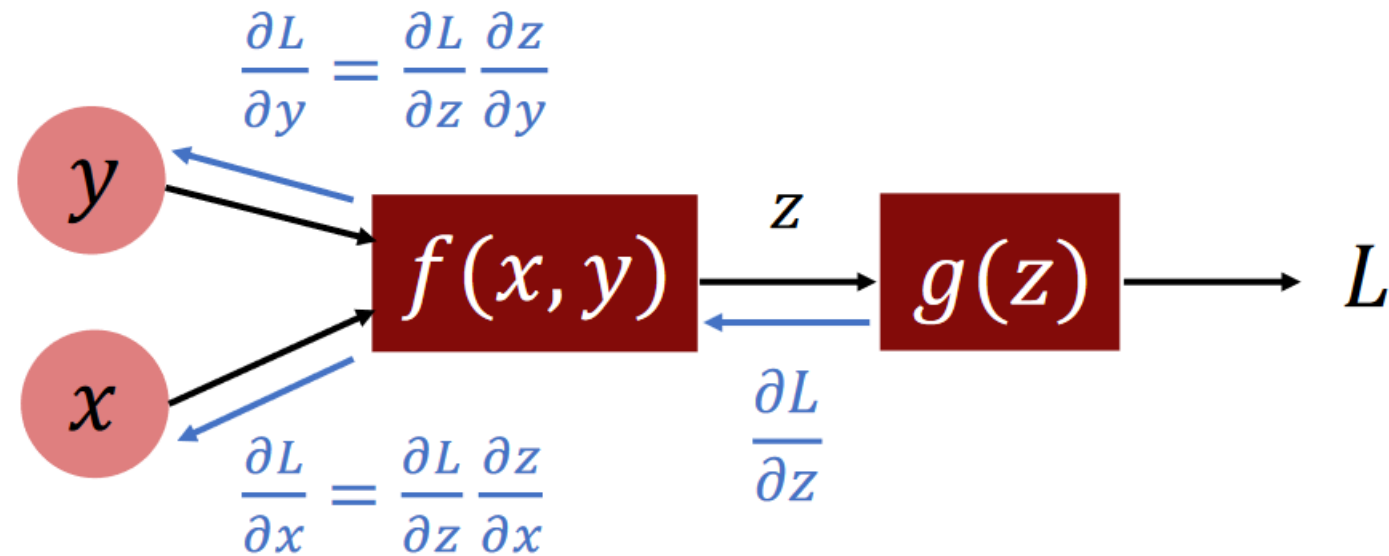
# Backpropagation



# Backpropagation

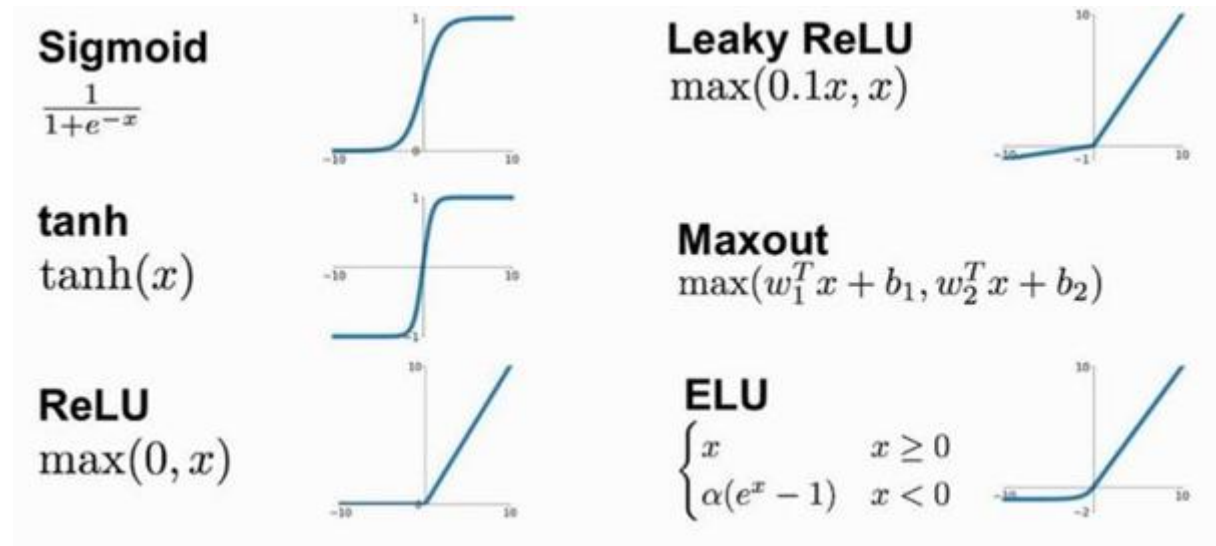


# Backpropagation



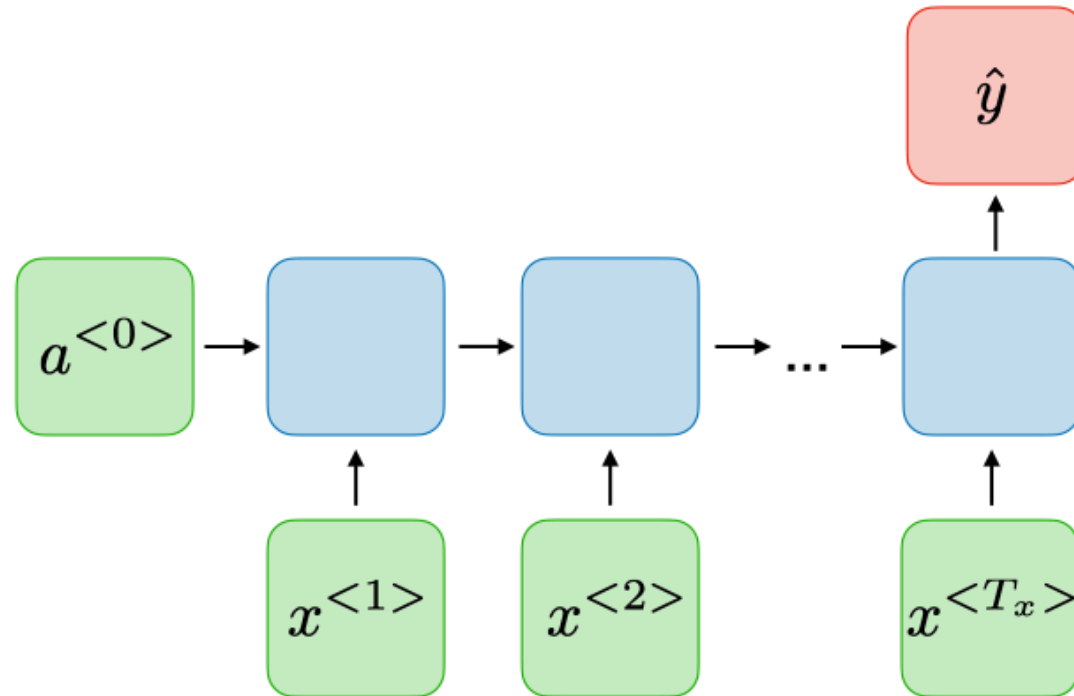
# Activation functions

$g$  should not be a linear function.



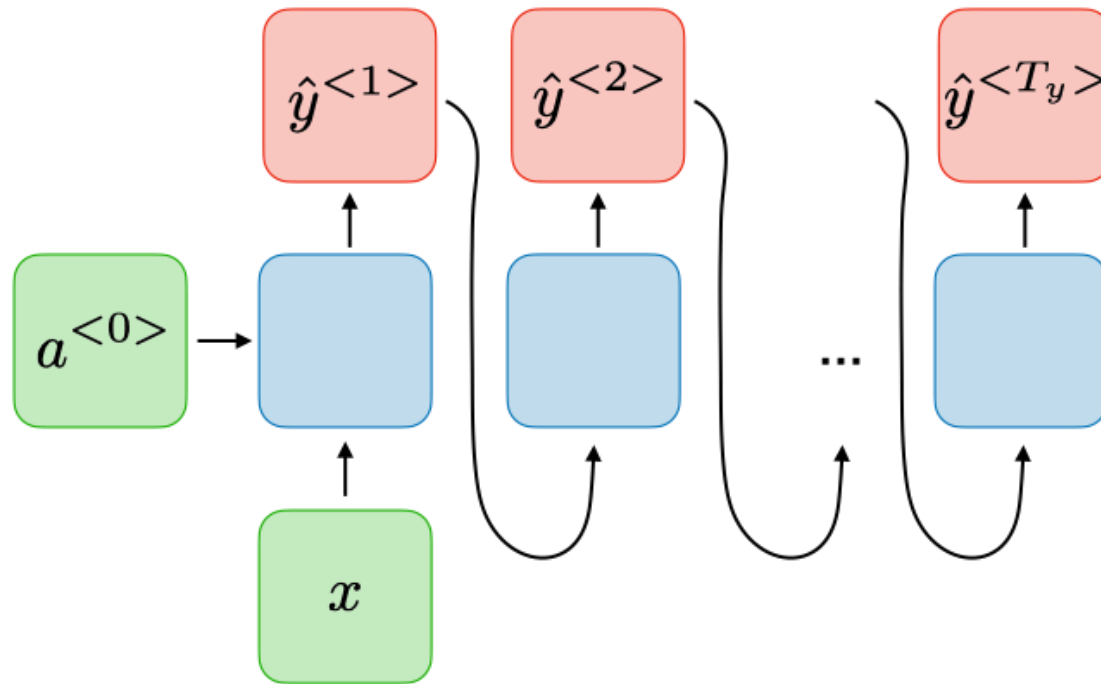
# Recurrent neural networks (RNN)

Many-to-one



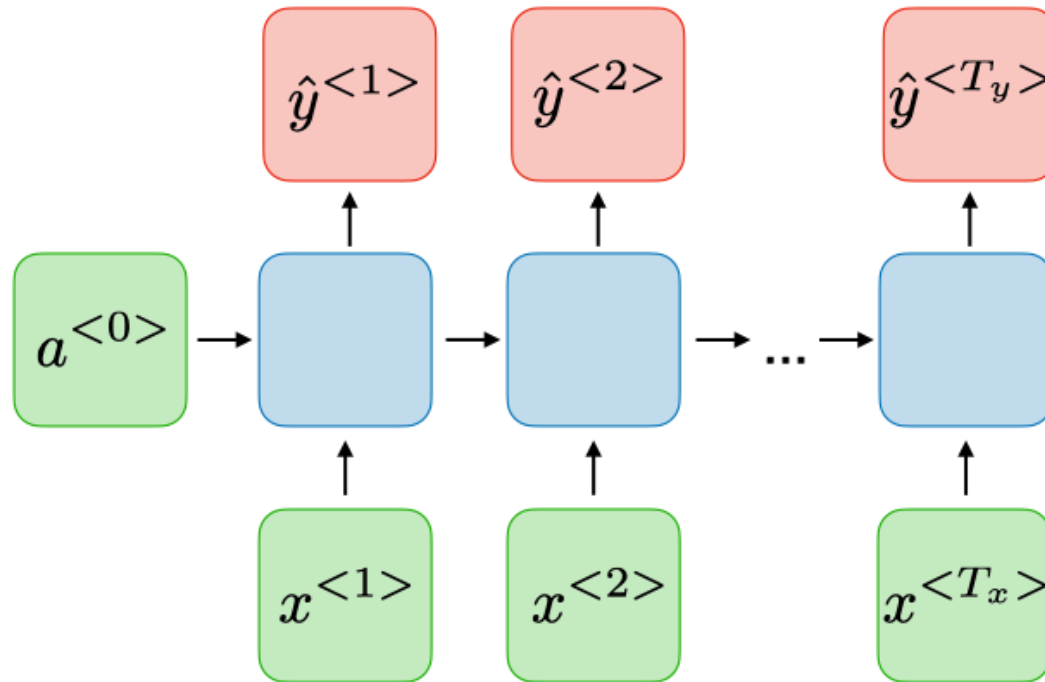
# Recurrent neural networks (RNN)

One-to-many



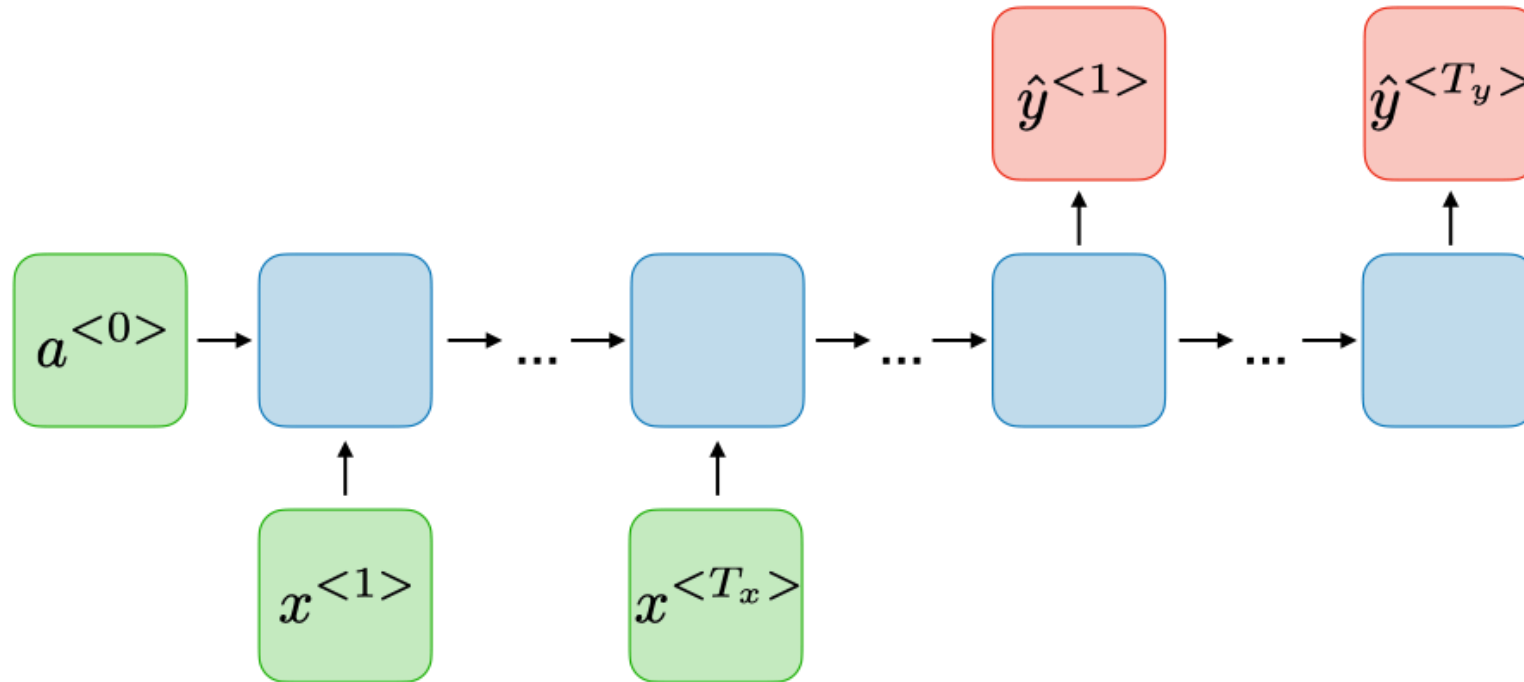
# Recurrent neural networks (RNN)

Many-to-many

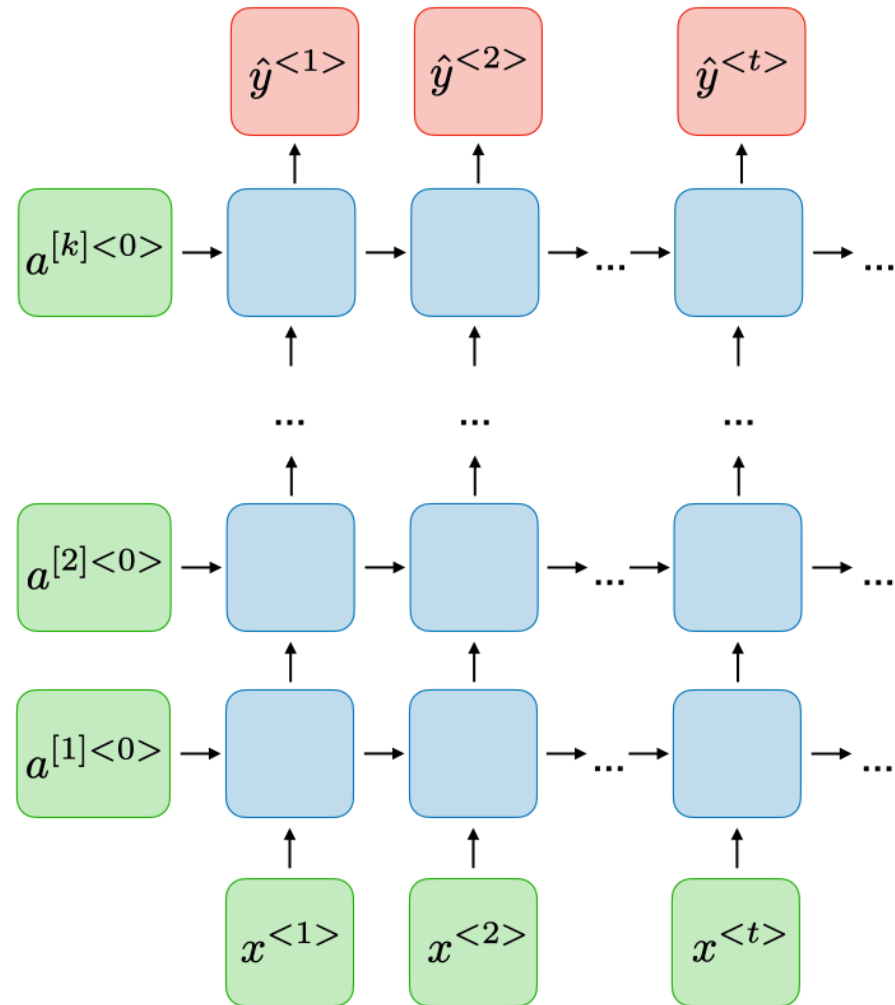


# Recurrent neural networks (RNN)

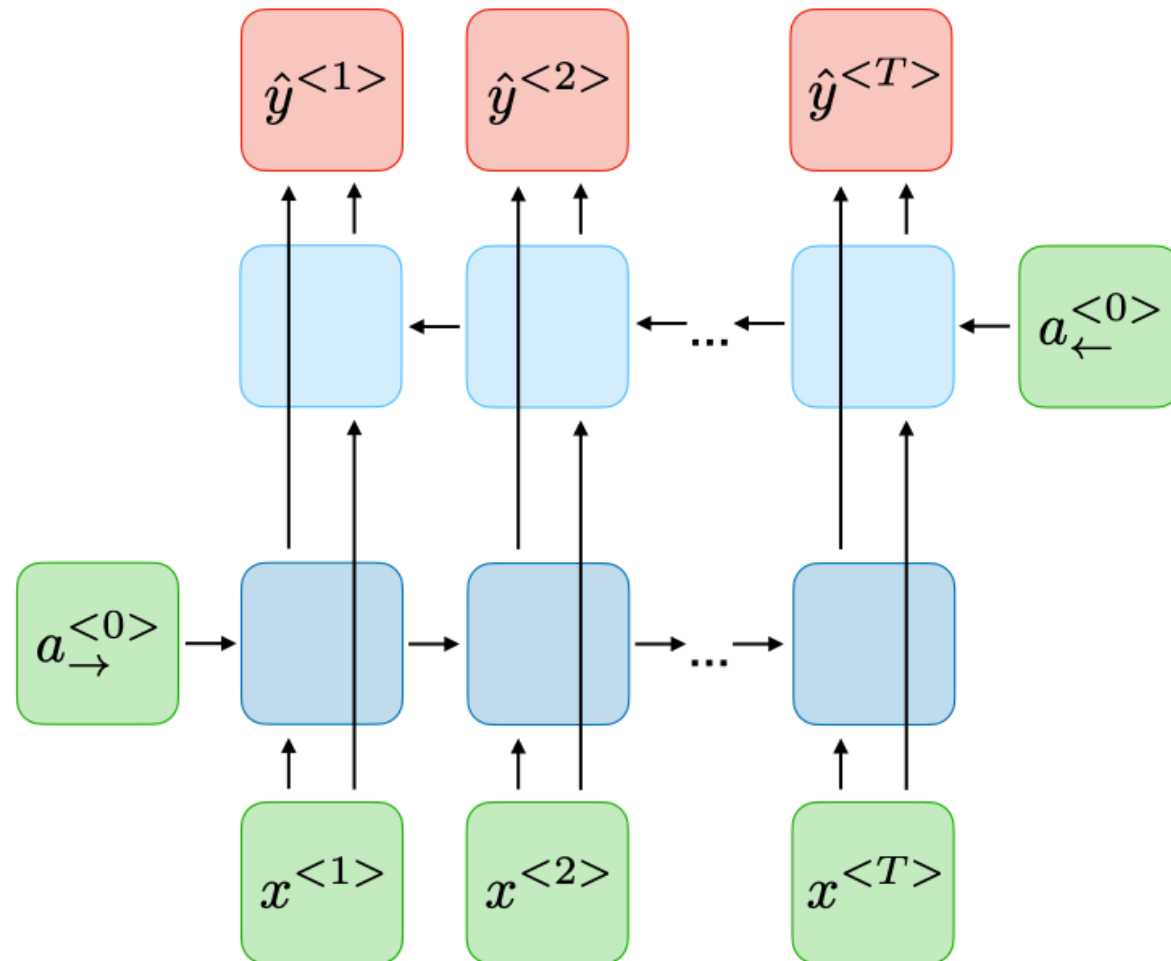
Many-to-many



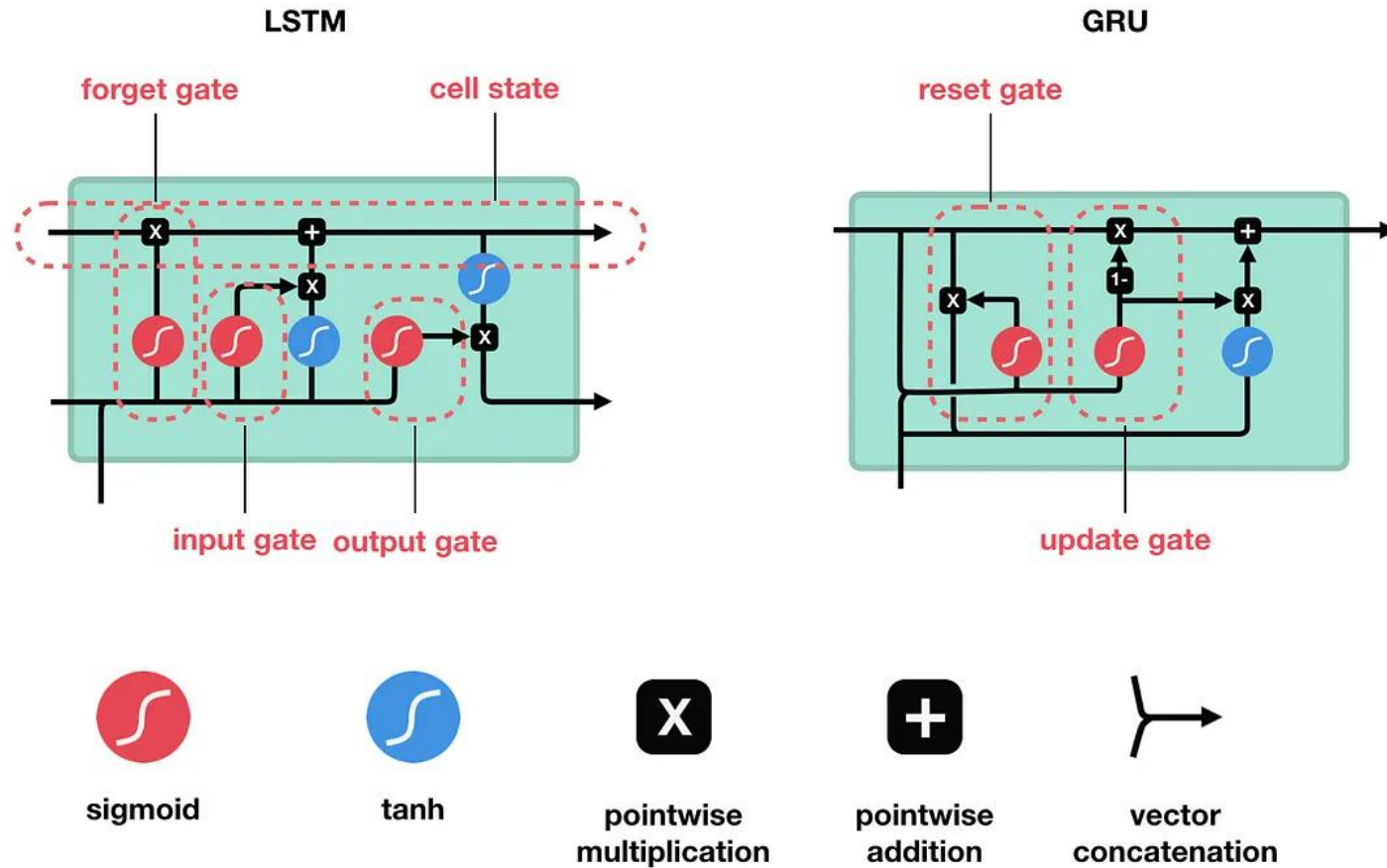
# Deep RNNs



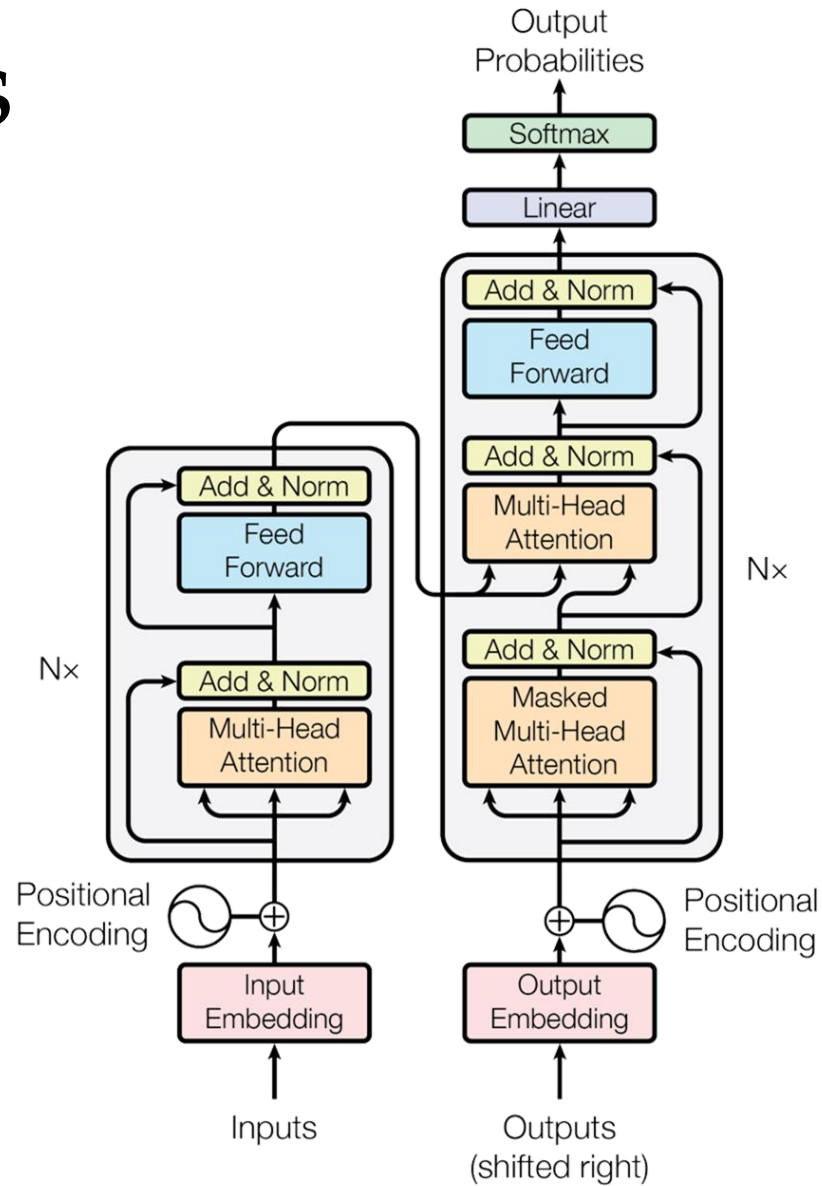
# Bidirectional RNNs



# LSTMs and GRUs



# Transformers



# Today...

- General course information
- Basics of robotics
- Fundamentals of machine learning

# Until next week...

Homework assignments will include programming with a machine learning library: PyTorch.

There are many online PyTorch tutorials. For what we covered today, check out:

- [https://pytorch.org/tutorials/beginner/blitz/tensor\\_tutorial.html](https://pytorch.org/tutorials/beginner/blitz/tensor_tutorial.html)
- [https://pytorch.org/tutorials/beginner/blitz/autograd\\_tutorial.html](https://pytorch.org/tutorials/beginner/blitz/autograd_tutorial.html)

# Next time...

- Basics of computer vision for robotics
- Representation learning